

(12) **United States Patent**
Memani et al.

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(54) **SYSTEM FOR MONITORING AND CLASSIFYING VEHICLE OPERATOR BEHAVIOR**

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(65) **Prior Publication Data**

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Related U.S. Application Data

(63) Continuation of application No. 16/288,884, filed on Feb. 28, 2019, now Pat. No. 11,667,299, which is a (Continued)

(51) **Int. Cl.**
B60W 40/09 (2012.01)
B60W 40/08 (2012.01)
(Continued)

(52) **U.S. Cl.**
CPC **B60W 40/09** (2013.01); **G06Q 10/1053** (2013.01); **G06Q 40/08** (2013.01);
(Continued)

(58) **Field of Classification Search**

CPC B60W 40/09; B60W 2040/0818; B60W 2040/0872; B60W 2556/45;

(Continued)

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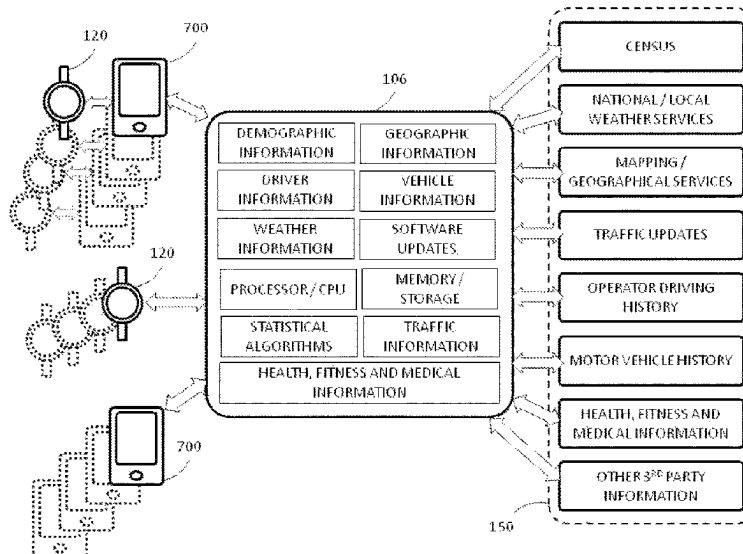
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Primary Examiner — Daniel M. Robert

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(57) **ABSTRACT**

System for determining qualification for insurance, cost of insurance or combinations thereof for a motor vehicle operator. At least one personal monitoring device monitors and communicates information about an operator of the motor vehicle, including at least driving data collected when the operator is operating the motor vehicle without receiving such driving data from vehicle sensors, calculated or derived based on data solely collected by the personal monitoring device without receiving such data from vehicle sensors, or combinations thereof. Application software on the device interfaces with the personal monitoring device hardware for monitoring and communicating, being activated and deactivated based on trigger events. At least one control system collects, processes and communicates at least a portion of the information collected and/or derived by the personal monitoring device used for determining qualification for insurance, cost of insurance or combinations thereof. At least one algorithm determines qualification for insurance, cost of insurance or combinations thereof for the operator based on at least a portion of the information collected by the personal monitoring device.

26 Claims, 31 Drawing Sheets



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(58) Field of Classification Search					705/14.66
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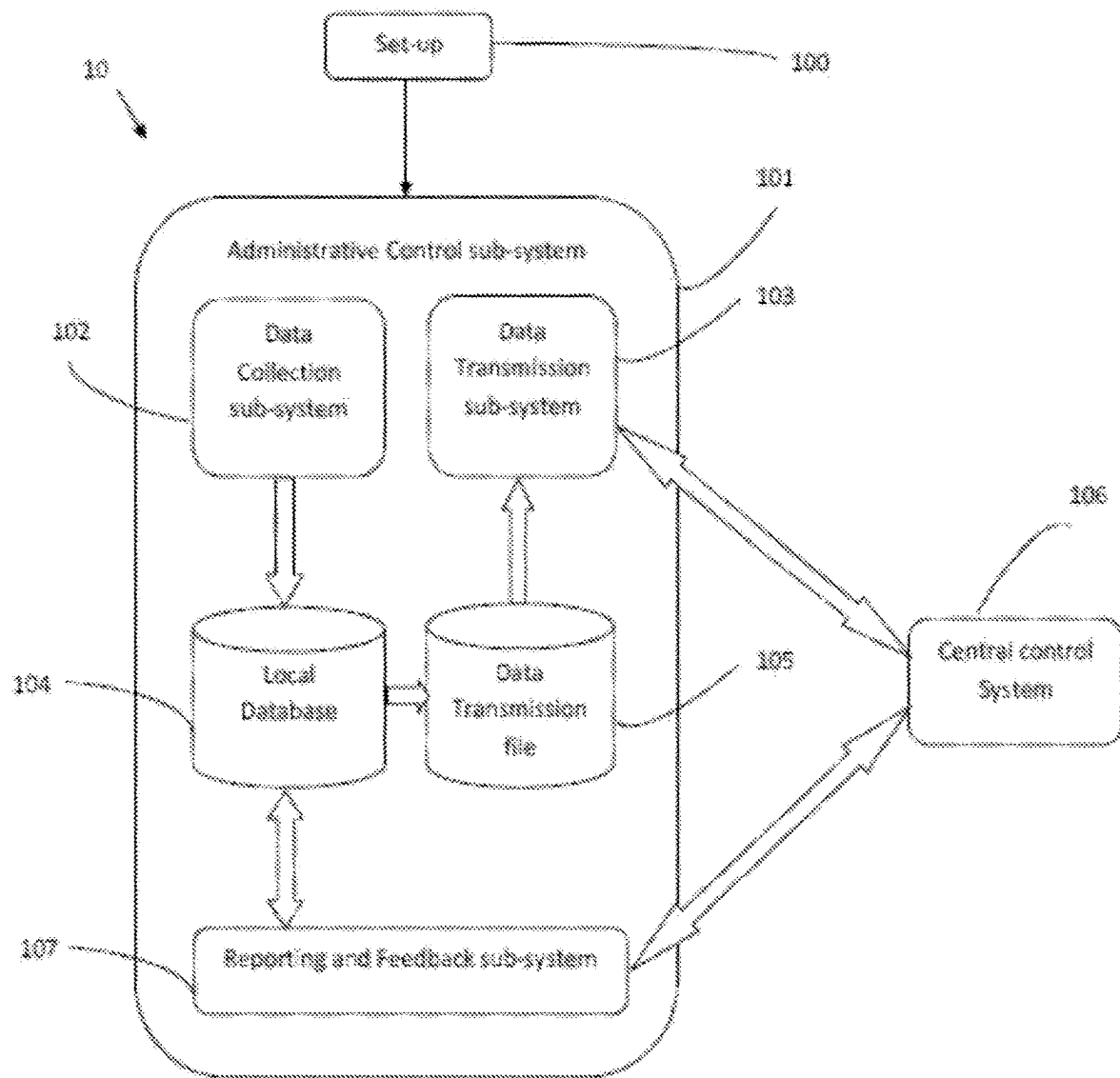


FIG. 1A

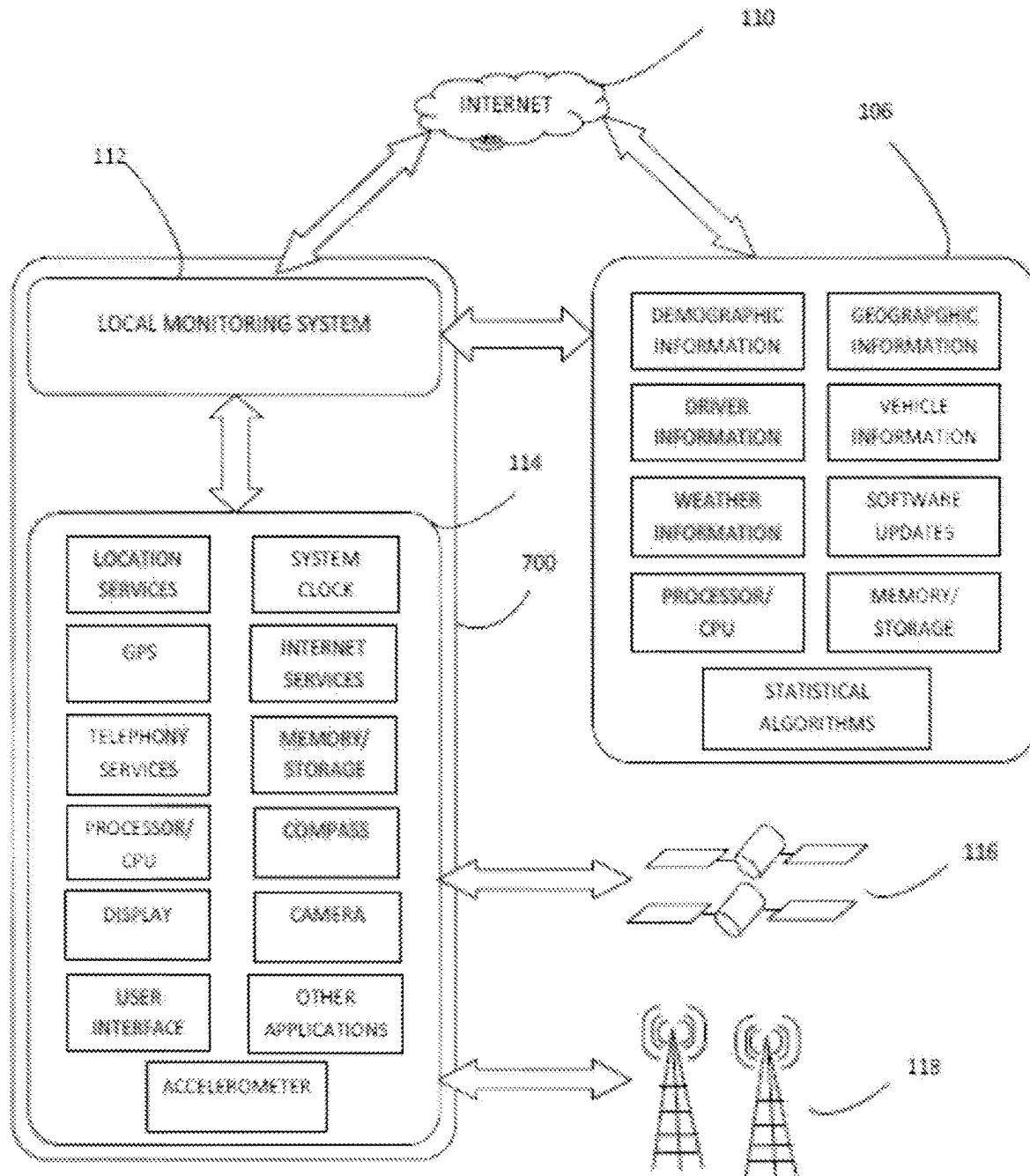


FIG. 1B

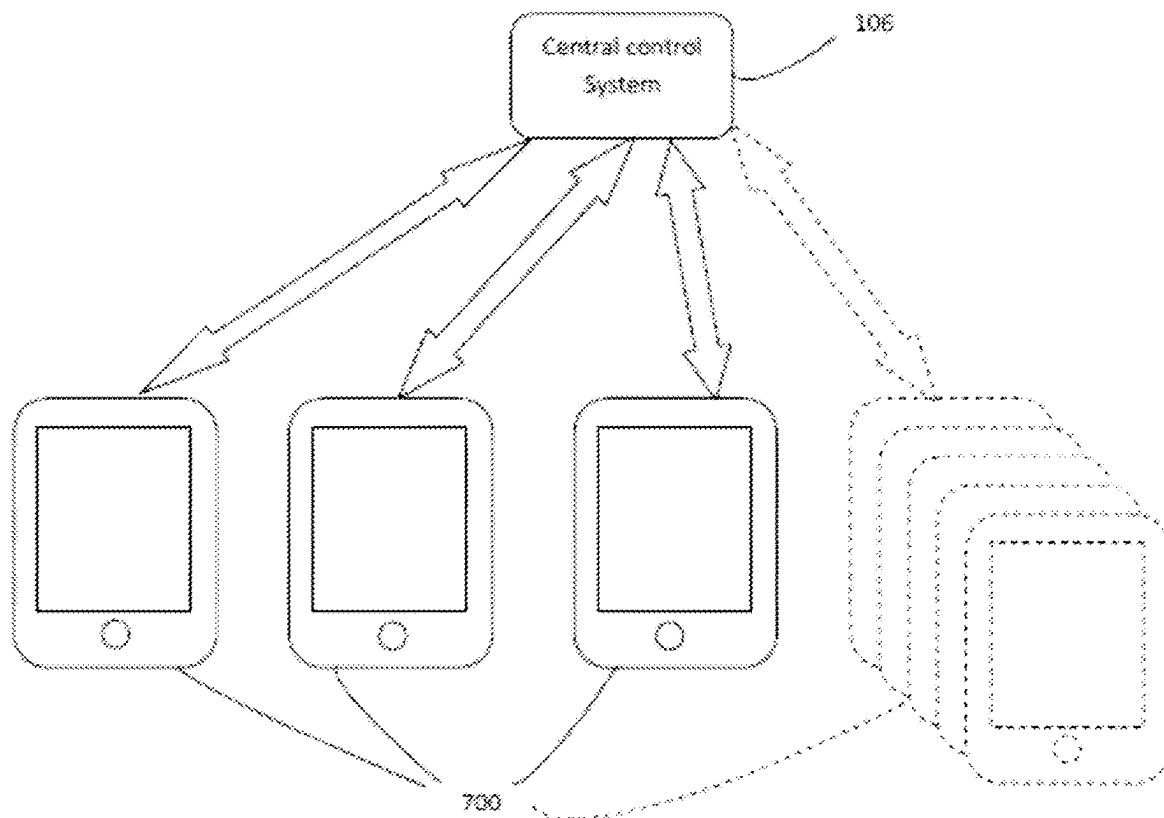


FIG. 1C

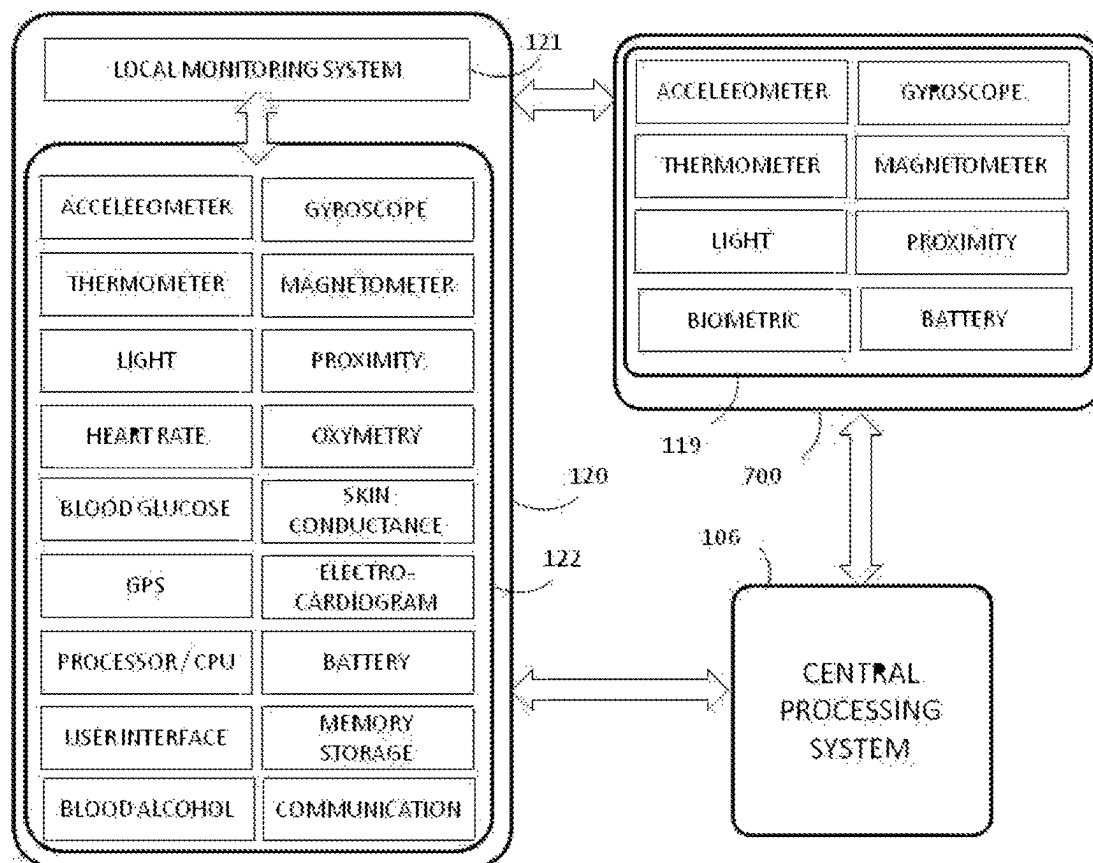


FIG. 1D

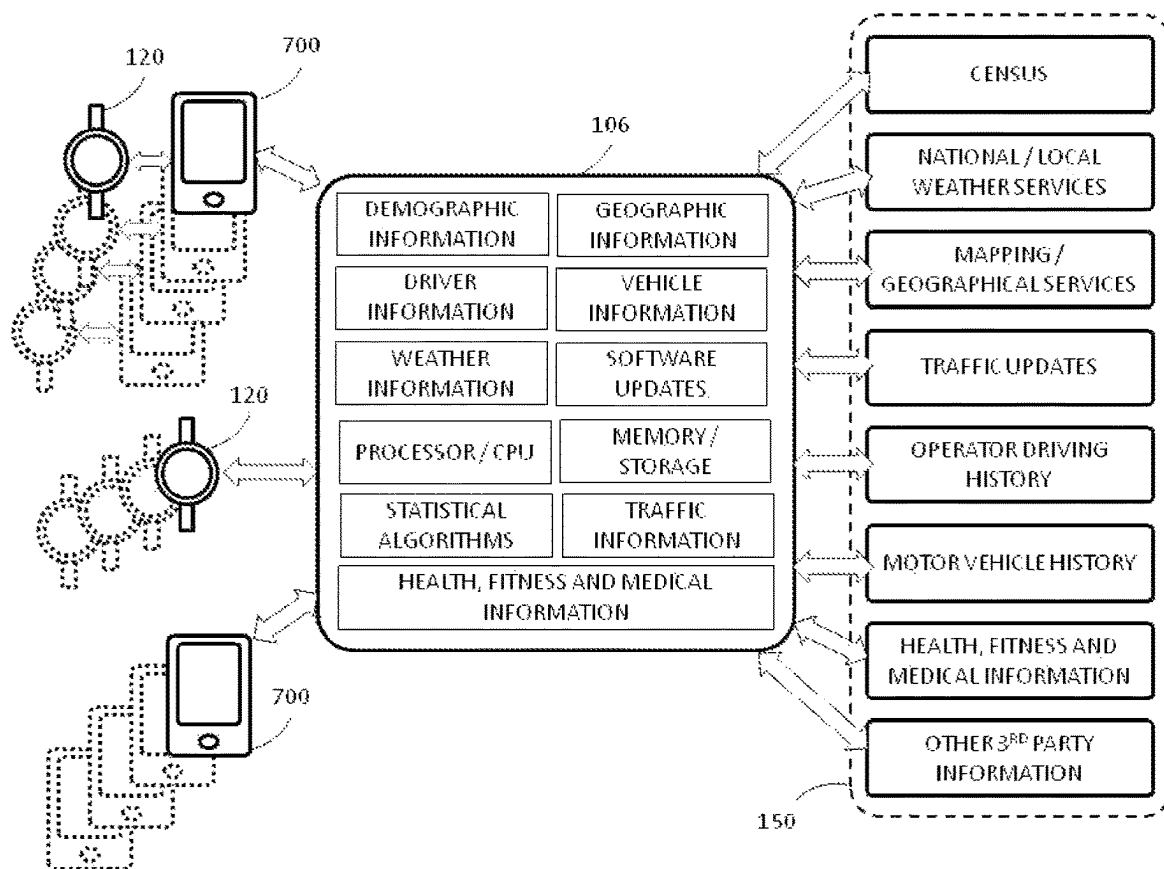


FIG. 1E

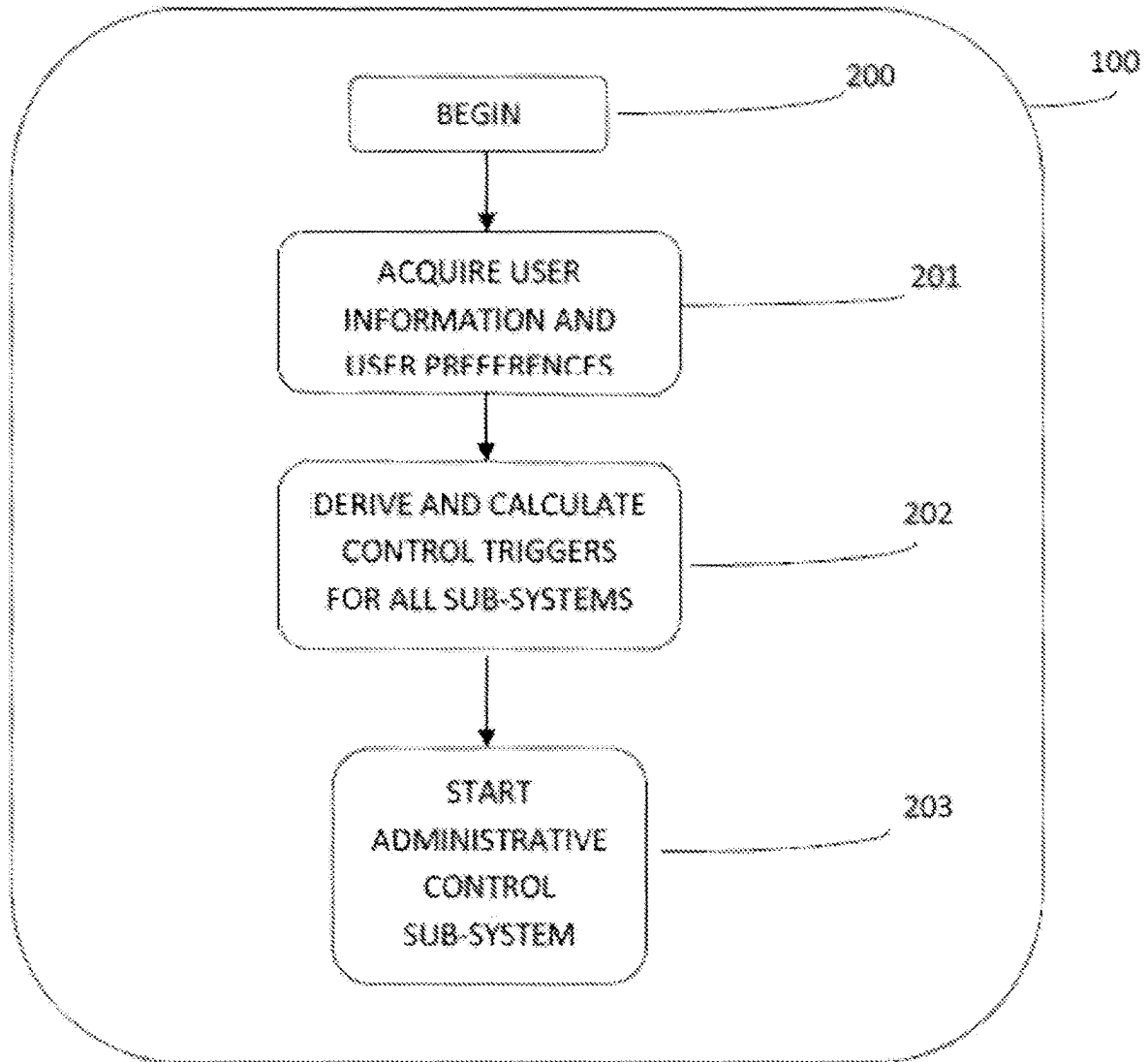


FIG. 2

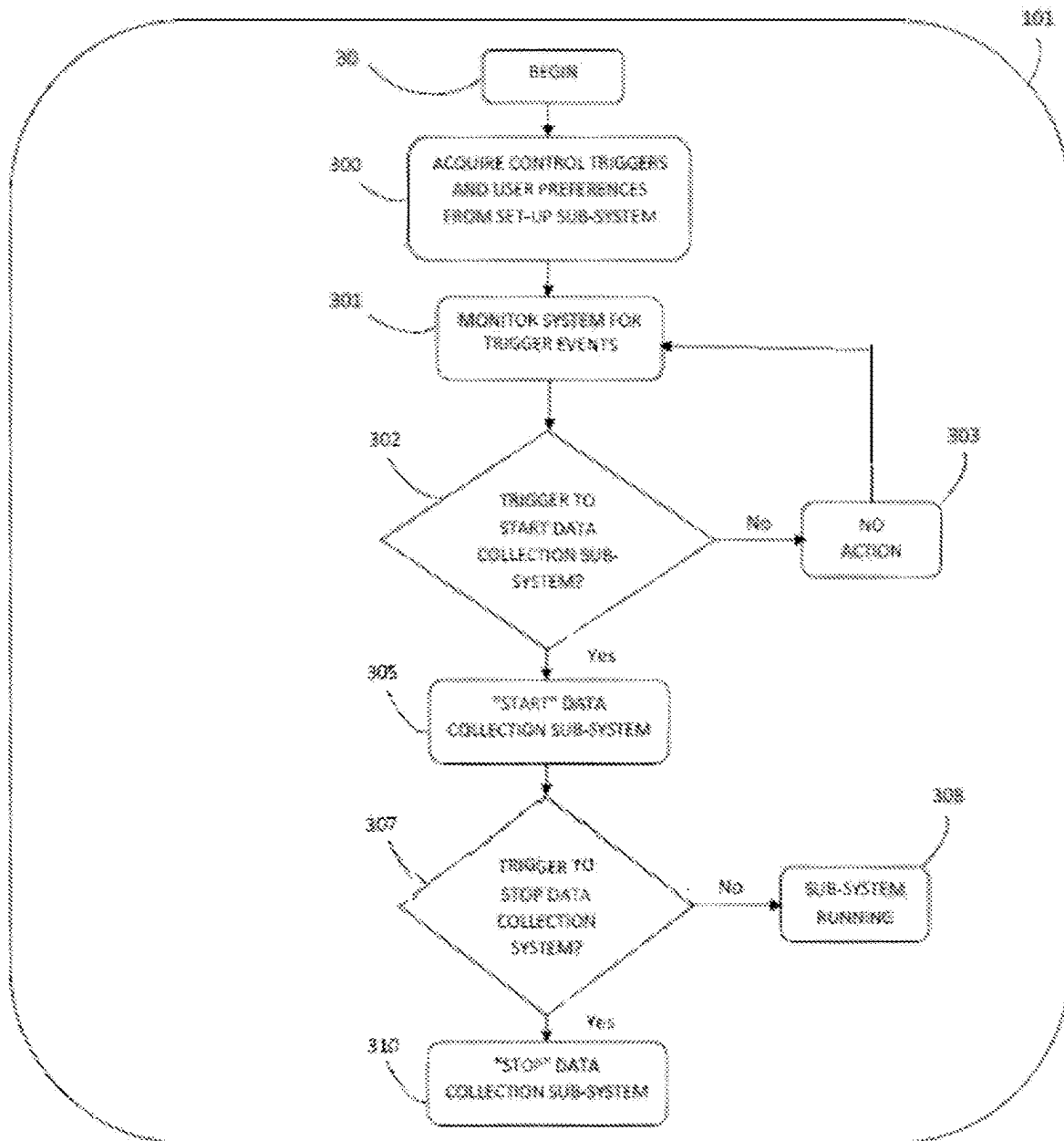


FIG. 3A

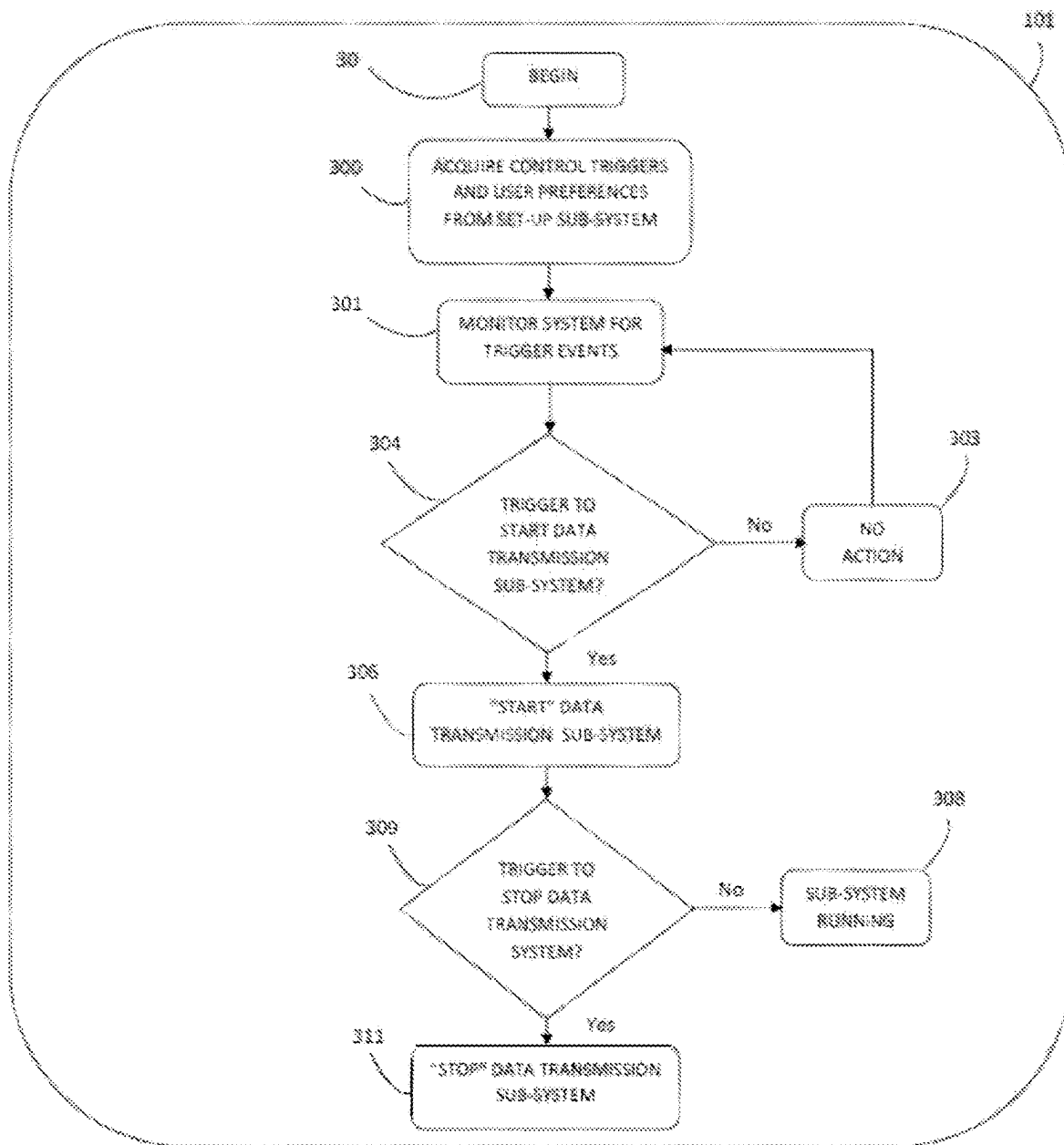


FIG. 3B

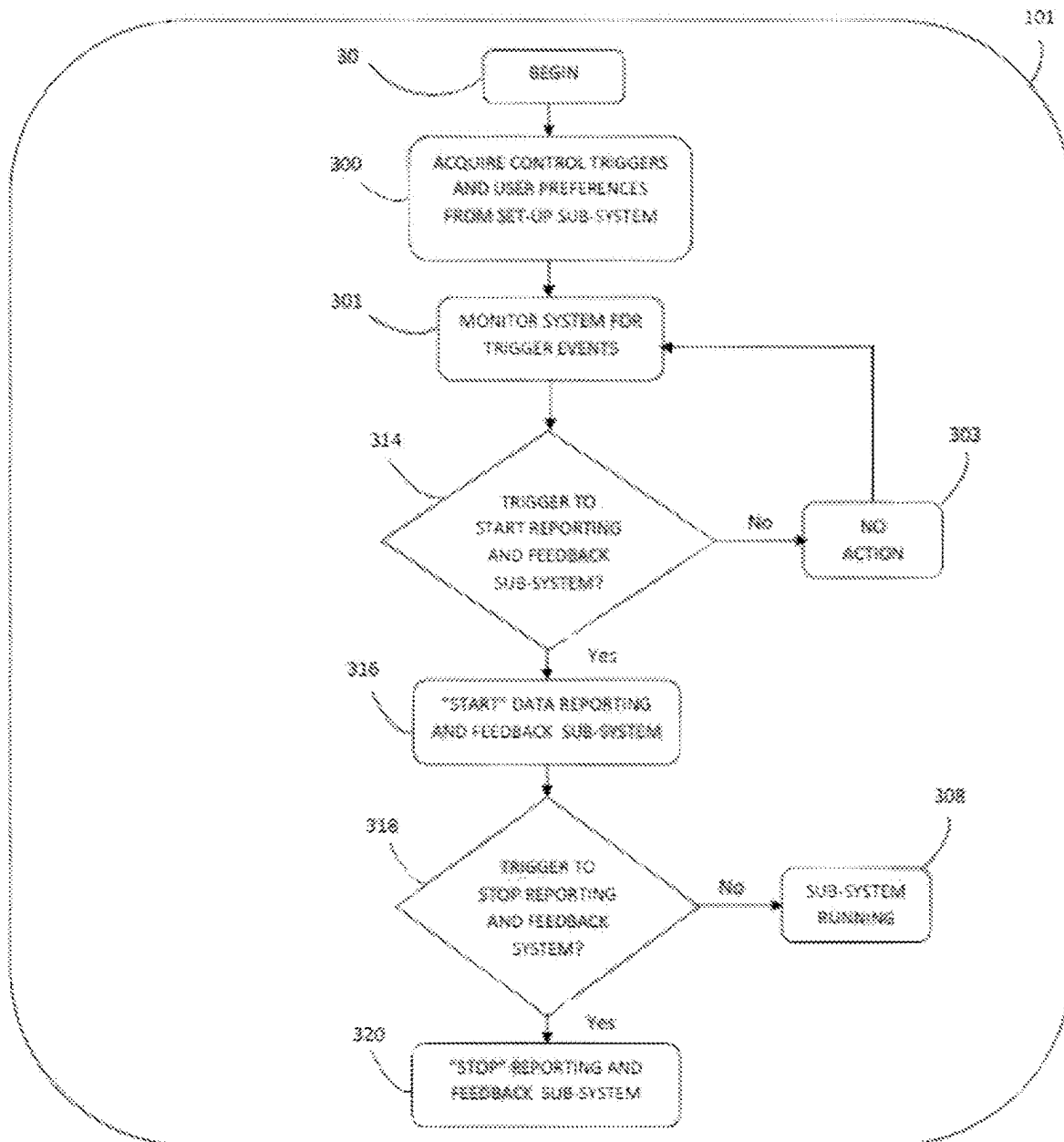


FIG. 3C

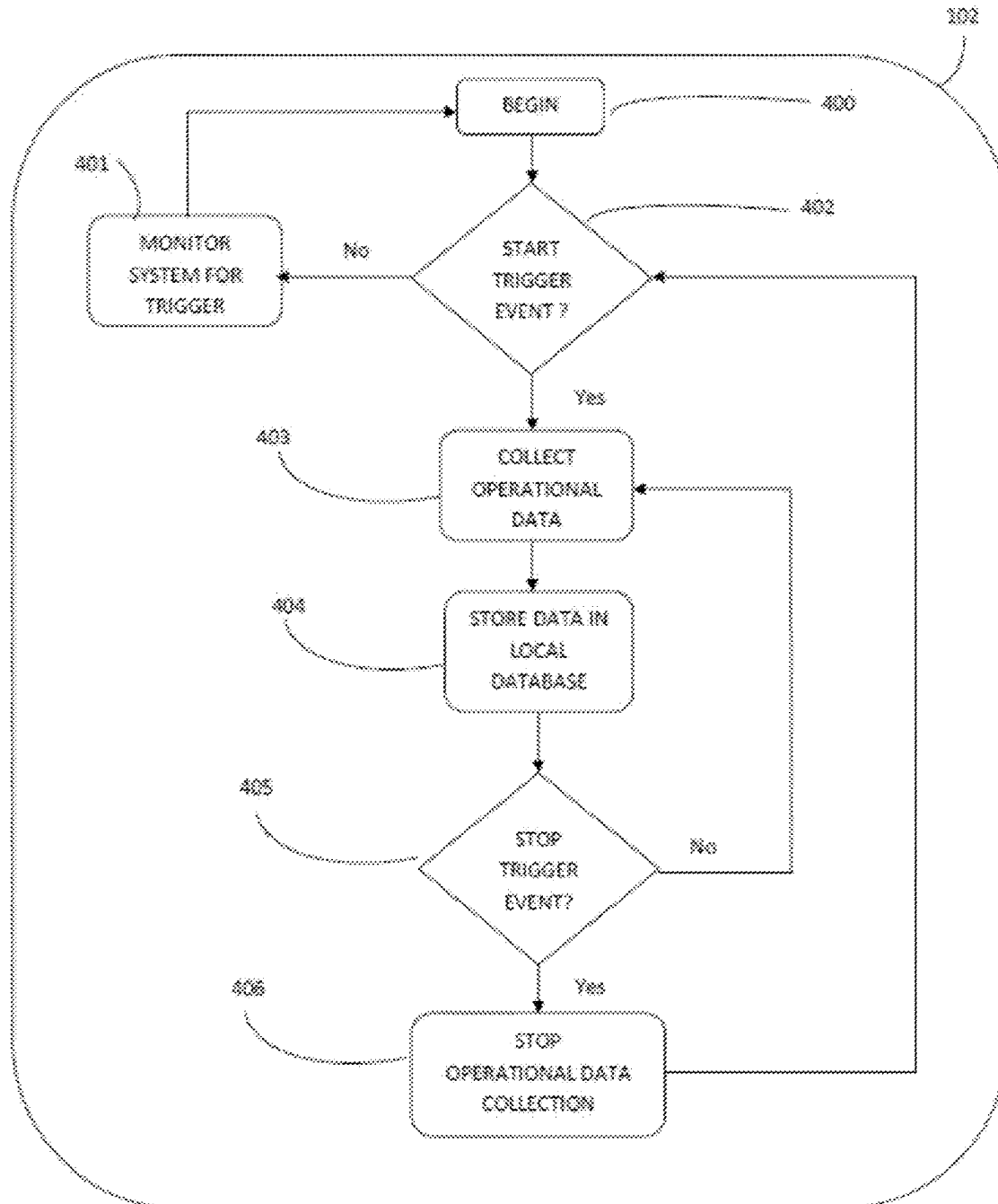


FIG. 4

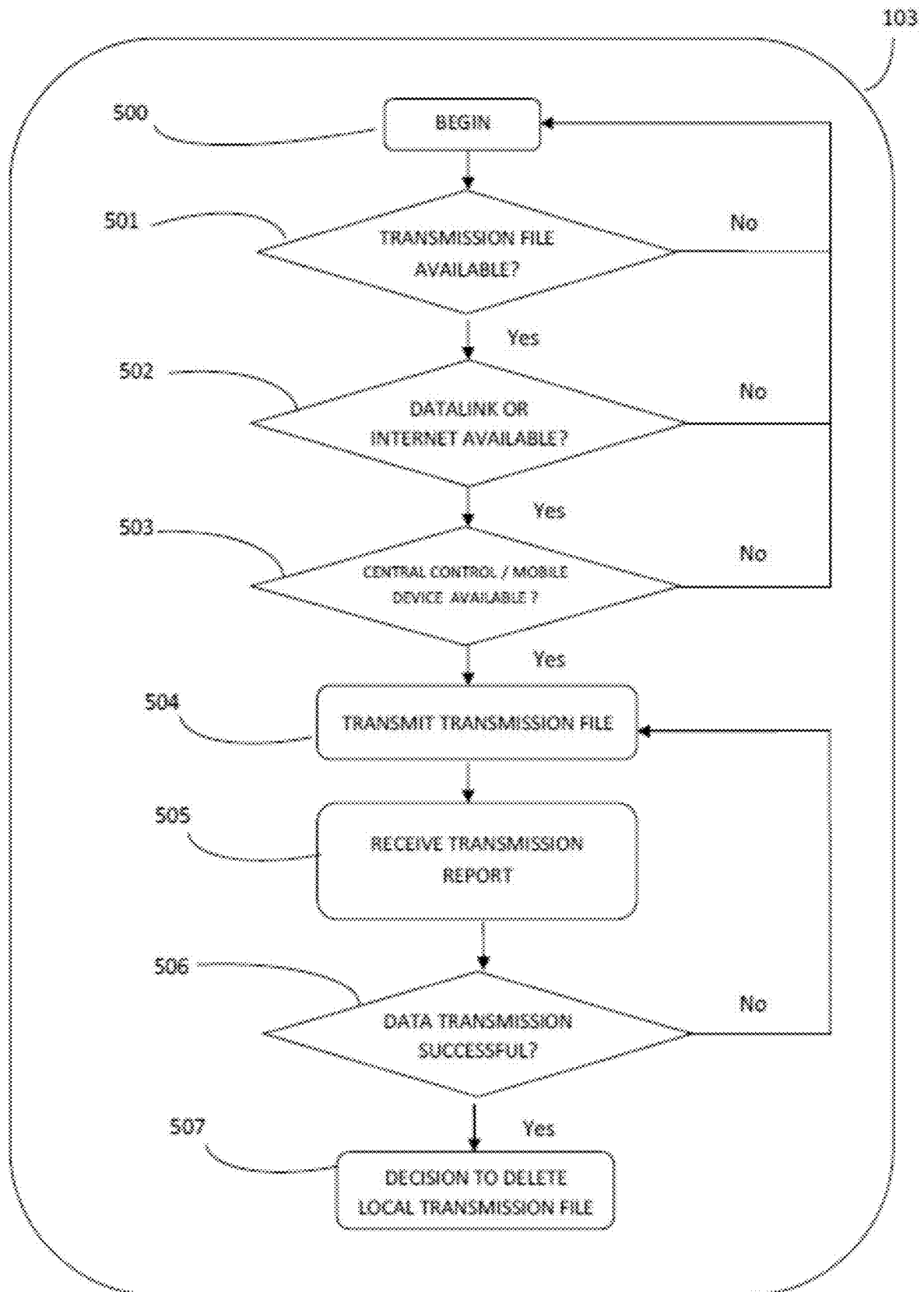


FIG 5

NAME OF THE SUB-SYSTEM	START TRIGGERS	STOP TRIGGERS	OTHER TRIGGERS
ADMINISTRATIVE CONTROL SUB-SYSTEM	SET-UP COMPLETE LICENSE RENEWED CONNECTION WITH MOBILE DEVICE FORCE START	END OF LICENSE TERM LOCAL DATABASE SIZE BATTERY CHARGE LOSS OF CONNECTION WITH MOBILE DEVICE FORCE STOP	SET-UP COMPLETE
DATA COLLECTION SUB-SYSTEM	CHANGE IN LOCATION SPEED TIME HEALTH AND FITNESS LEVEL CHANGE IN MEDICAL CONDITION PRESENCE OF DRUG OR ALCOHOL LIGHT CONDITIONS FORCE START	SPEED TIME ALTITUDE HEALTH AND FITNESS LEVEL CHANGE IN MEDICAL CONDITION PRESENCE OF DRUG OR ALCOHOL LIGHT CONDITIONS FORCE STOP	DEMOGRAPHIC PROFILE AGE GENDER HEIGHT WEIGHT MEDICAL PROFILE AND HISTORY
DATA TRANSMISSION SUB-SYSTEM	PRESENCE OF LOCAL DATA PRESENCE OF MOBILE DEVICE FILE SIZE TIME FORCE START	TIME NUMBER OF ATTEMPTS LOSS OF CONNECTION WITH MOBILE DEVICE FORCE STOP	TYPE OF CONNECTIVITY WITH MOBILE DEVICE

FIG. 6

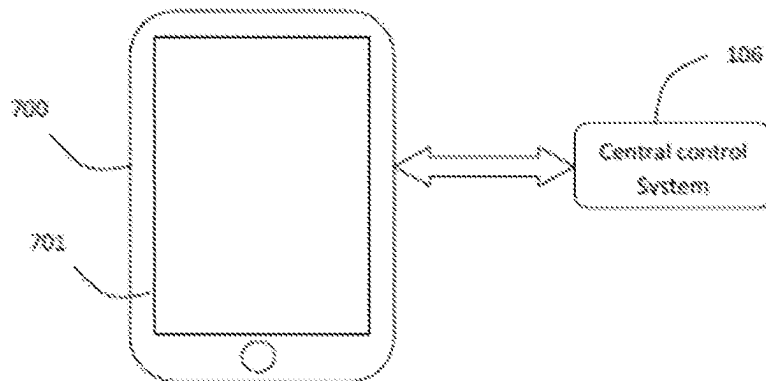


FIG. 7A

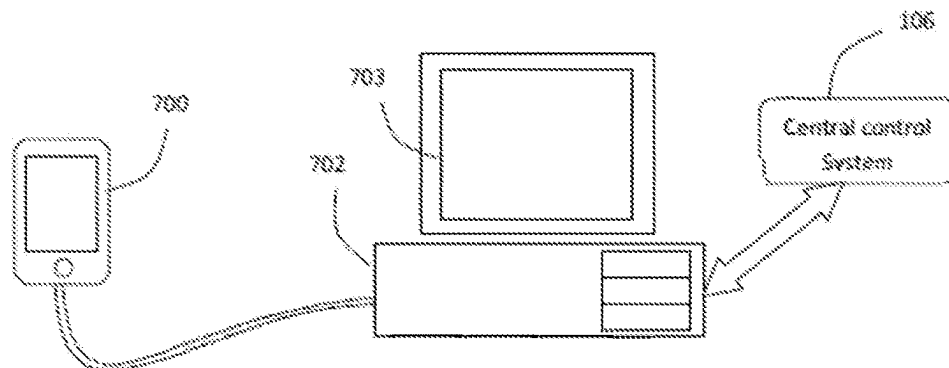


FIG. 7B

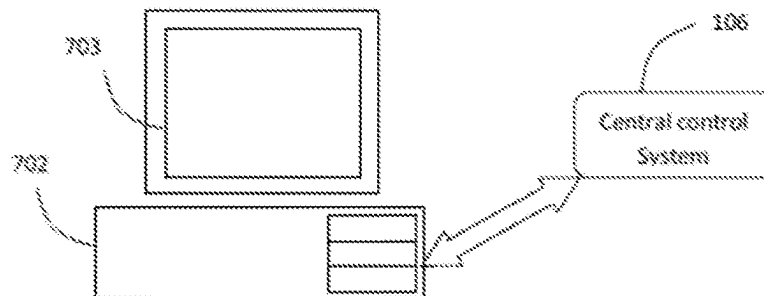


FIG. 7C

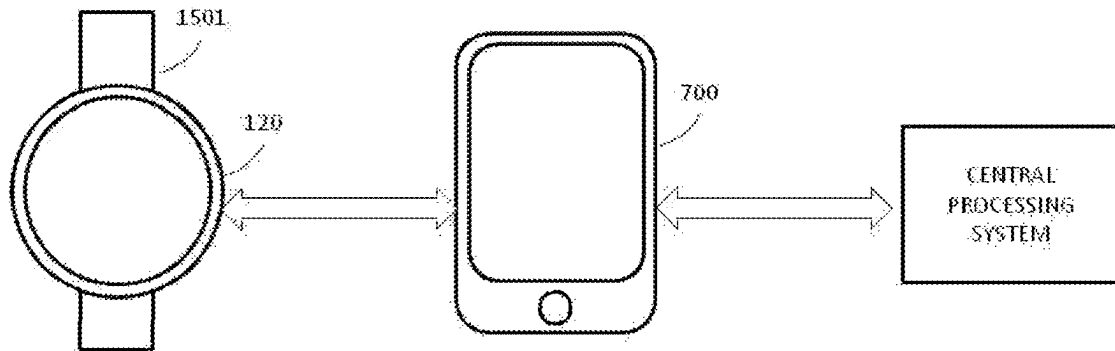


FIG. 7D

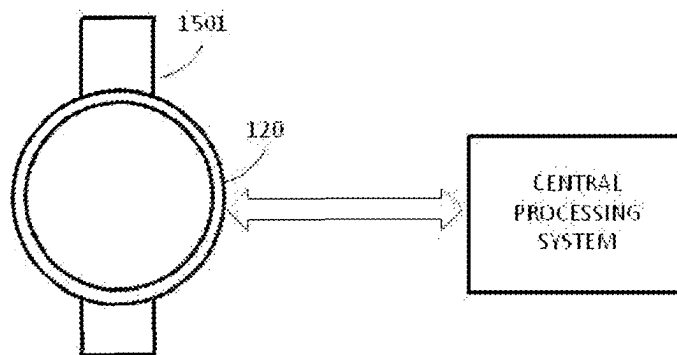


FIG. 7E

Customer Number	Date	Time	Speed	Speed	Zip	Direction	Use of Cell		Latitude	Longitude	Step	Active time
				Limit	code		Phone				count	
ABC123456789DM	8/7/2015	18:41:01	28.1	25.0	43016	280.0	Idle		40.11	-83.16	6,500	0:22:47
ABC123456789DM	8/7/2015	18:41:03	29.6	25.0	43016	279.1	Idle		40.11	-83.16	6,500	0:22:47
ABC123456789DM	8/7/2015	18:41:04	30.8	25.0	43016	279.6	Idle		40.11	-83.16	6,500	0:22:47
ABC123456789DM	8/7/2015	18:41:34	30.4	25.0	43016	341.5	Idle		40.11	-83.16	6,500	0:22:47
ABC123456789DM	8/7/2015	18:42:25	29.3	25.0	43016	40.6	Idle		40.12	-83.16	6,500	0:22:47
ABC123456789DM	8/7/2015	18:43:10	22.6	25.0	43016	264.2	Idle		40.12	-83.16	6,500	0:22:47
ABC123456789DM	8/7/2015	18:43:21	14.6	25.0	43016	8.2	Idle		40.12	-83.16	6,500	0:22:47
ABC123456789DM	8/7/2015	18:43:22	17.1	25.0	43016	11.0	Idle		40.12	-83.16	6,500	0:22:47
ABC123456789DM	8/7/2015	18:43:24	19.4	25.0	43016	5.7	Idle		40.12	-83.16	6,500	0:22:47
ABC123456789DM	8/7/2015	18:43:27	22.3	25.0	43016	355.9	Idle		40.12	-83.16	6,500	0:22:47
ABC123456789DM	8/7/2015	18:43:37	27.0	25.0	43016	316.6	Idle		40.12	-83.16	6,500	0:22:47
ABC123456789DM	8/7/2015	18:43:42	27.0	25.0	43016	316.6	Telephone Call		40.12	-83.16	6,500	0:22:47
ABC123456789DM	8/7/2015	18:44:12	22.8	25.0	43016	352.8	Telephone Call		40.11	-83.16	6,500	0:22:47
ABC123456789DM	8/7/2015	18:45:03	19.2	25.0	43016	337.4	Telephone Call		40.11	-83.16	6,500	0:22:47
ABC123456789DM	8/7/2015	18:45:08	20.9	25.0	43016	335.2	Telephone Call		40.11	-83.16	6,500	0:22:47
ABC123456789DM	8/7/2015	18:45:13	22.1	25.0	43016	340.0	Telephone Call		40.11	-83.16	6,500	0:22:47
ABC123456789DM	8/7/2015	18:45:18	23.4	25.0	43016	344.3	Telephone Call		40.11	-83.16	6,500	0:22:47
ABC123456789DM	8/7/2015	18:45:23	27.0	25.0	43016	351.5	Telephone Call		40.11	-83.16	6,500	0:22:47
ABC123456789DM	8/7/2015	18:45:29	29.0	25.0	43016	353.0	Telephone Call		40.11	-83.16	6,500	0:22:47
ABC123456789DM	8/7/2015	18:45:34	30.3	25.0	43016	354.6	Telephone Call		40.11	-83.16	6,500	0:22:47
ABC123456789DM	8/7/2015	18:45:44	32.1	25.0	43016	355.0	Telephone Call		40.11	-83.16	6,500	0:22:47
ABC123456789DM	8/7/2015	18:45:49	36.1	25.0	43016	359.5	Telephone Call		40.11	-83.16	6,500	0:22:47
ABC123456789DM	8/7/2015	18:45:59	39.1	25.0	43016	6.5	Telephone Call		40.11	-83.16	6,500	0:22:47
ABC123456789DM	8/7/2015	18:47:10	17.0	25.0	43016	309.3	Telephone Call		40.11	-83.16	6,500	0:22:47
ABC123456789DM	8/7/2015	18:47:16	18.8	25.0	43016	306.5	Telephone Call		40.11	-83.16	6,500	0:22:47
ABC123456789DM	8/7/2015	18:47:21	20.1	25.0	43016	307.6	Telephone Call		40.11	-83.16	6,500	0:22:47
ABC123456789DM	8/7/2015	18:47:26	21.4	25.0	43016	301.6	Telephone Call		40.11	-83.16	6,500	0:22:47
ABC123456789DM	8/7/2015	18:47:31	22.8	25.0	43016	295.6	Telephone Call		40.11	-83.16	6,500	0:22:47
ABC123456789DM	8/7/2015	18:47:36	24.6	25.0	43016	290.3	Telephone Call		40.11	-83.16	6,500	0:22:47
ABC123456789DM	8/7/2015	18:47:46	26.0	25.0	43016	284.1	Telephone Call		40.11	-83.16	6,500	0:22:47
ABC123456789DM	8/7/2015	18:47:52	28.1	25.0	43016	280.0	Telephone Call		40.11	-83.16	6,500	0:22:47
ABC123456789DM	8/7/2015	18:48:02	29.6	25.0	43016	279.1	Idle		40.11	-83.16	6,500	0:22:47
ABC123456789DM	8/7/2015	18:48:02	30.8	25.0	43016	279.6	Idle		40.11	-83.16	6,500	0:22:47

FIG. 8A

Customer Number	Date	Time	Blood Pressure	Blood Glucose	Body Temperature	Heart Rate	Blood Alcohol
ABC123456789DM	8/2/2015	0:00:00	122/76	90	98.00	68	0.00
ABC123456789DM	8/2/2015	2:00:00	122/76	90	97.80	64	0.00
ABC123456789DM	8/2/2015	4:00:00	122/76	88	97.50	62	0.00
A8C123456789DM	8/2/2015	6:00:23	122/76	87	97.50	84	0.00
A8C123456789DM	8/2/2015	8:01:00	125/77	85	98.80	78	0.00
A8C123456789DM	8/2/2015	10:00:01	125/77	100	99.00	71	0.00
A8C123456789DM	8/2/2015	12:00:00	122/76	125	99.40	72	0.00
A8C123456789DM	8/2/2015	14:01:11	120/80	160	99.40	72	0.00
A8C123456789DM	8/2/2015	16:00:15	120/80	140	99.60	71	0.01
A8C123456789DM	8/2/2015	18:00:00	120/80	125	99.30	71	0.02
A8C123456789DM	8/2/2015	20:00:00	120/74	155	98.80	74	0.02
A8C123456789DM	8/2/2015	22:00:48	120/74	140	98.00	73	0.01
A8C123456789DM	8/3/2015	0:00:01	122/76	100	97.80	69	0.00
A8C123456789DM	8/3/2015	2:00:22	122/76	85	97.50	63	0.00
A8C123456789DM	8/3/2015	4:00:00	122/76	80	97.50	60	0.00
A8C123456789DM	8/3/2015	6:01:02	121/75	80	98.20	82	0.00
A8C123456789DM	8/3/2015	8:01:07	121/75	95	98.60	77	0.00
A8C123456789DM	8/3/2015	10:00:00	121/75	120	99.10	72	0.00
A8C123456789DM	8/3/2015	12:00:00	123/76	135	99.30	72	0.00
A8C123456789DM	8/3/2015	14:00:01	123/76	165	99.40	71	0.00
A8C123456789DM	8/3/2015	16:00:15	122/76	144	99.50	73	0.00
A8C123456789DM	8/3/2015	18:00:37	122/78	151	99.40	74	0.00
A8C123456789DM	8/3/2015	20:01:02	123/77	135	98.60	74	0.00
A8C123456789DM	8/3/2015	22:02:00	121/75	110	98.10	73	0.00
A8C123456789DM	8/4/2015	0:00:00	121/75	90	97.90	68	0.00
A8C123456789DM	8/4/2015	2:00:00	121/75	91	97.40	62	0.00
A8C123456789DM	8/4/2015	4:00:47	121/75	88	97.20	60	0.00
A8C123456789DM	8/4/2015	6:00:23	122/76	95	98.00	81	0.00
A8C123456789DM	8/4/2015	8:00:07	122/76	118	98.80	75	0.00
A8C123456789DM	8/4/2015	10:00:01	120/74	115	99.00	72	0.00
A8C123456789DM	8/4/2015	12:00:00	120/74	130	99.50	71	0.00
A8C123456789DM	8/4/2015	14:01:00	120/74	166	99.30	73	0.00
A8C123456789DM	8/4/2015	16:01:00	121/75	141	99.70	72	0.00
A8C123456789DM	8/4/2015	18:00:25	121/75	135	99.40	74	0.03
A8C123456789DM	8/4/2015	20:00:05	122/76	165	98.70	75	0.01
A8C123456789DM	8/4/2015	22:00:12	122/76	138	98.10	74	0.00

FIG. 8B

Customer Number	Date	Time	Body Temperature	Awake time	Active time	Walking Steps	Running Steps	Stair Steps	Calories	Miles
ABC123456789DM	8/2/2015	0:00:00	98.00	0:00:00	0:00:00	0	0	0	0	0.0
ABC123456789DM	8/2/2015	2:00:00	97.80	0:00:00	0:00:00	0	0	0	0	0.0
ABC123456789DM	8/2/2015	4:00:00	97.50	0:00:00	0:00:00	0	0	0	0	0.0
ABC123456789DM	8/2/2015	6:00:23	97.50	0:57:32	0:32:00	4002	5115	0	373	4.6
ABC123456789DM	8/2/2015	8:01:00	98.80	2:57:32	0:48:00	4280	5115	2614	448	6.0
ABC123456789DM	8/2/2015	10:00:01	99.00	4:57:32	0:48:22	4301	5115	2614	448	6.0
ABC123456789DM	8/2/2015	12:00:00	99.40	6:57:32	0:48:37	4345	5115	2614	449	6.0
ABC123456789DM	8/2/2015	14:01:11	99.40	8:57:32	0:52:00	4680	5115	2614	458	6.2
ABC123456789DM	8/2/2015	16:00:15	99.60	10:57:32	0:52:00	4695	5115	2614	459	6.2
ABC123456789DM	8/2/2015	18:00:00	99.30	12:57:32	1:02:11	4842	5115	2845	468	6.4
ABC123456789DM	8/2/2015	20:00:00	98.80	14:38:21	1:07:00	5078	5115	2845	475	6.5
ABC123456789DM	8/2/2015	22:00:48	98.00	14:38:21	1:12:00	5402	5115	2912	485	6.7
ABC123456789DM	8/3/2015	0:00:01	97.80	0:00:00	0:00:00	0	0	0	0	0.0
ABC123456789DM	8/3/2015	2:00:22	97.50	0:00:00	0:00:00	0	0	0	0	0.0
ABC123456789DM	8/3/2015	4:00:00	97.50	0:00:00	0:00:00	0	0	0	0	0.0
ABC123456789DM	8/3/2015	6:01:02	98.20	0:36:14	0:00:00	0	0	0	0	0.0
ABC123456789DM	8/3/2015	8:01:07	98.60	2:36:14	0:35:00	3800	5265	2459	439	5.8
ABC123456789DM	8/3/2015	10:00:00	99.10	4:36:14	0:52:17	4412	5265	2459	455	6.1
ABC123456789DM	8/3/2015	12:00:00	99.30	6:36:14	0:54:00	4526	5265	2459	458	6.1
ABC123456789DM	8/3/2015	14:00:01	99.40	8:36:14	1:00:25	4701	5265	2502	464	6.2
ABC123456789DM	8/3/2015	16:00:15	99.50	10:36:14	1:02:00	4786	5265	2502	466	6.3
ABC123456789DM	8/3/2015	18:00:37	99.40	12:36:14	1:07:17	5214	5265	2502	477	6.5
ABC123456789DM	8/3/2015	20:01:02	98.60	14:27:32	1:12:29	5589	5265	2578	489	6.7
ABC123456789DM	8/3/2015	22:02:00	98.10	15:02:39	1:17:55	5644	5265	2607	491	6.8
ABC123456789DM	8/4/2015	0:00:00	97.90	0:00:00	0:00:00	0	0	0	0	0.0
ABC123456789DM	8/4/2015	2:00:00	97.40	0:00:00	0:00:00	0	0	0	0	0.0
ABC123456789DM	8/4/2015	4:00:47	97.20	0:00:00	0:00:00	0	0	0	0	0.0
ABC123456789DM	8/4/2015	6:00:23	98.00	0:41:02	0:00:00	158	0	0	4	0.1
ABC123456789DM	8/4/2015	8:00:07	98.80	2:41:02	0:29:00	4865	2675	2848	341	5.2
ABC123456789DM	8/4/2015	10:00:01	99.00	4:41:02	0:42:55	4925	4680	2848	448	6.2
ABC123456789DM	8/4/2015	12:00:00	99.50	6:41:02	0:48:12	4865	4680	2914	448	6.2
ABC123456789DM	8/4/2015	14:01:00	99.30	8:41:02	0:54:35	5412	4680	3017	465	6.6
ABC123456789DM	8/4/2015	16:01:00	99.70	10:41:02	0:57:24	5489	4845	3017	476	6.7
ABC123456789DM	8/4/2015	18:00:25	99.40	12:41:02	1:12:00	5674	4845	3017	480	6.8
ABC123456789DM	8/4/2015	20:00:05	98.70	14:41:02	1:14:27	5715	4845	3026	482	6.8
ABC123456789DM	8/4/2015	22:00:12	98.10	16:55:19	1:15:39	5802	4845	3026	484	6.8

FIG. 8C

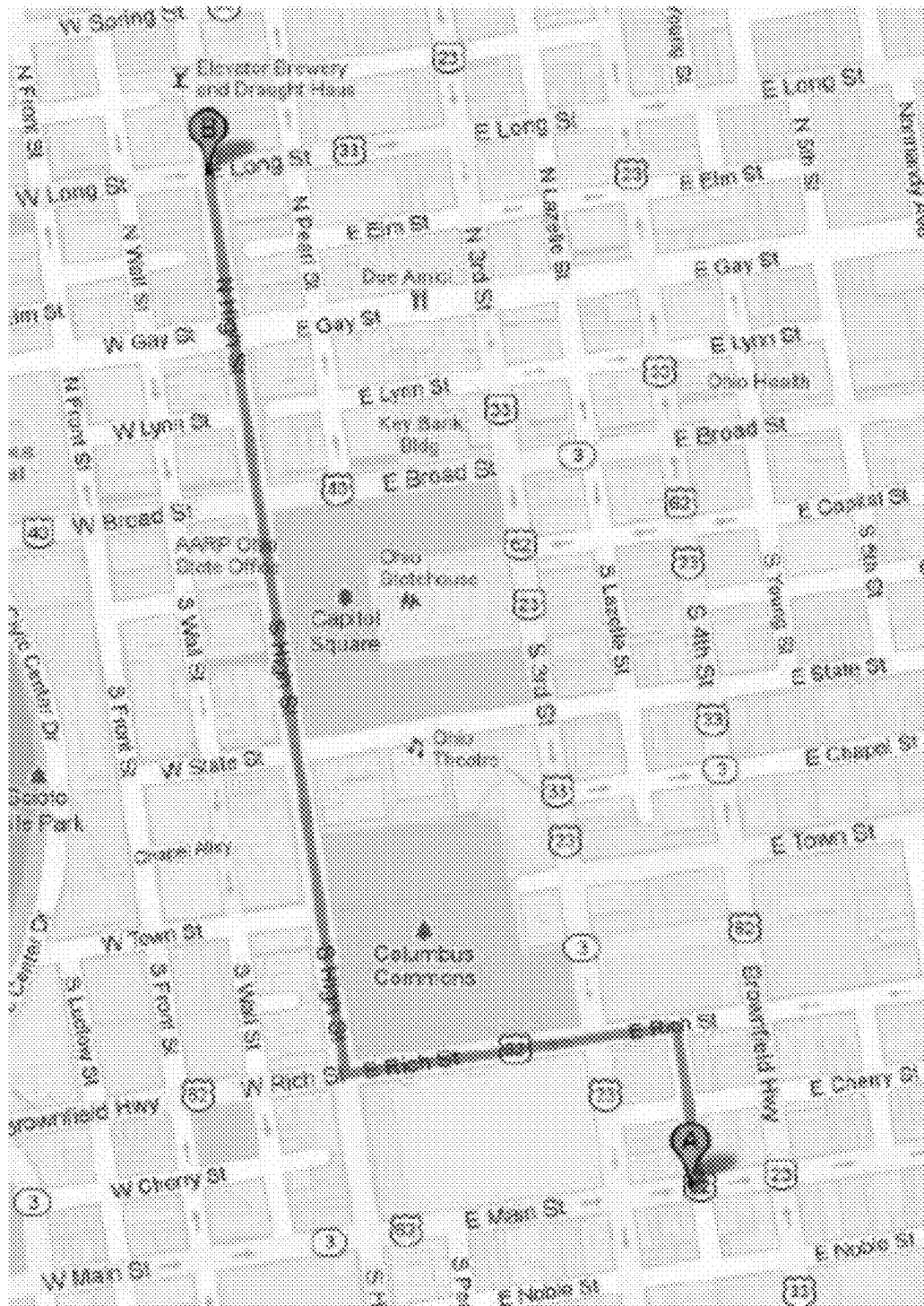


FIG. 9

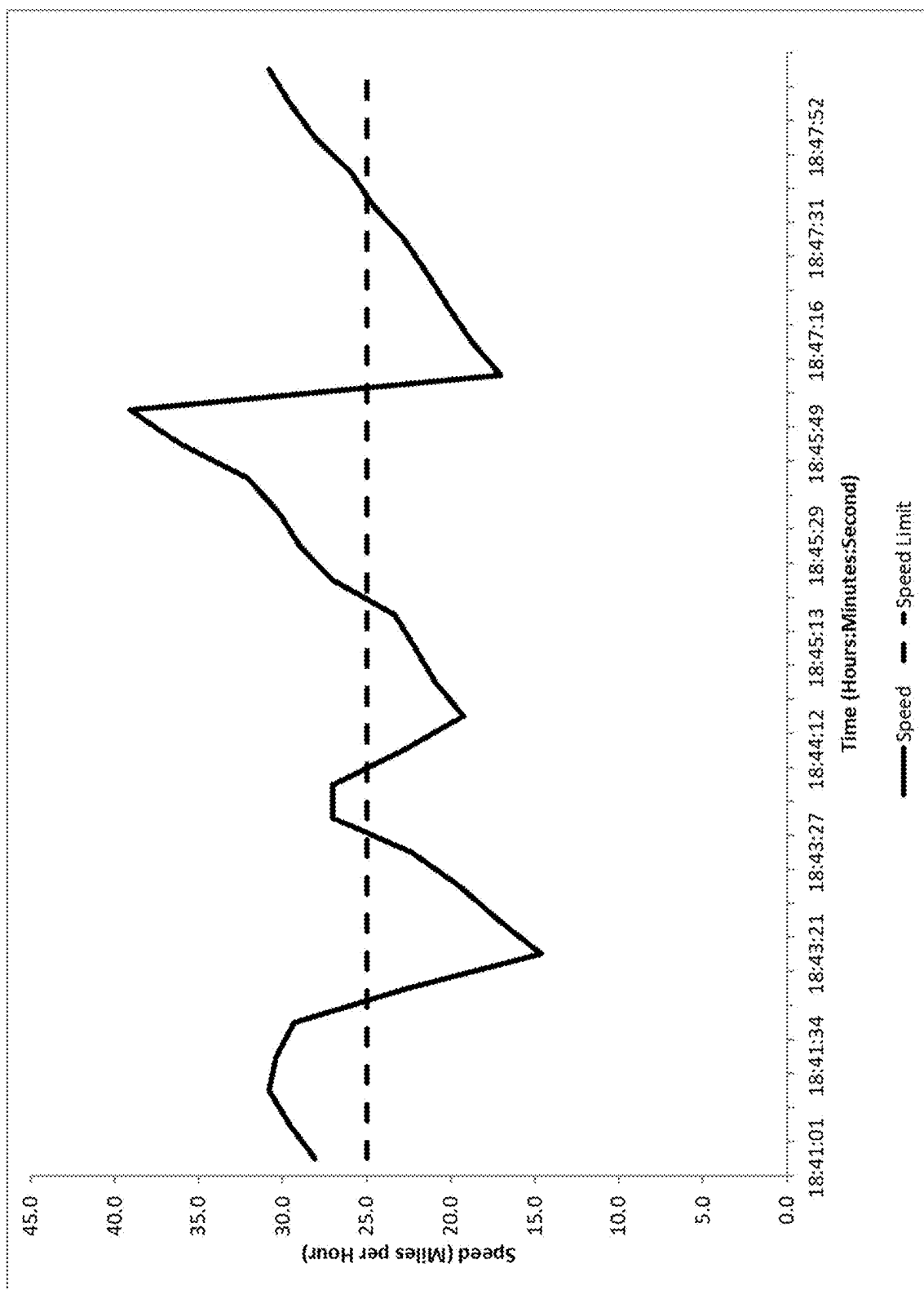


FIG. 10

REPORT PERIOD		
Start date	May 1 2015	
End date	May 31 2015	
CUSTOMER NUMBER		
ABC123456789DM		
		CLASSIFICATION
	CUSTOMER	GROUP
TOTAL PERFORMANCE	45	4th Decile
DETAILED PERFORMANCE		
Operator	65	6th Decile
Vehicle	30	3rd Decile
Geography	50	5th Decile
Environment	35	3th Decile

FIG. 11A

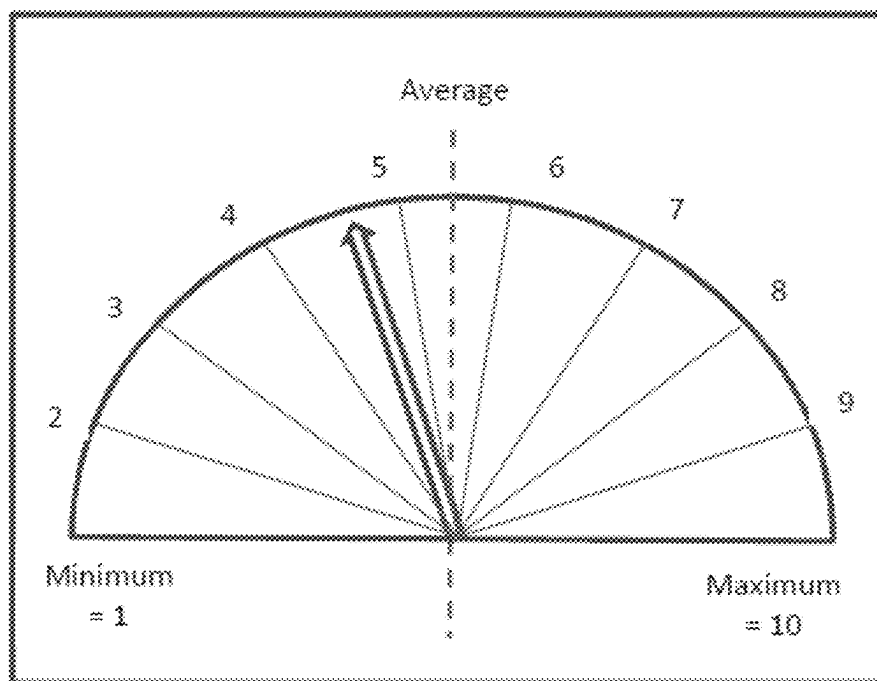


FIG. 11B

	Operator Behavior	
	CUSTOMER	CLASSIFICATION GROUP
TOTAL OPERATOR BEHAVIOR	65	4th Decile
DRIVING BEHAVIOR		
Distance travelled	4000 miles	6th Decile
Distance at Night	800 miles	3rd Decile
Distance during Day	1200 miles	5th Decile
Distance during Rush Hours	2000 miles	3th Decile
Geographies visited	9	8th Decile
Average Speed	48 mph	4th Decile
Roads <35mph	26 pmh	2nd Decile
Roads with Speed 35-50mph	44 mph	3rd Decile
Roads with Speed 50+mph	67 mph	9th Decile
Speed above legal speed limit	7 times	5th Decile
Phone use while driving	14%	10th Decile
Rapid accelerations	4 times	4th Decile
Hard Brakings	12 times	5th Decile
PHYSICAL FITNESS		
Active Time / Idle Time	0.8	4th Decile
Exercise	0.45 hours	8th Decile
Fitness level	4	6th Decile
Alcohol / drug use	0	1st Decile
CAPACITY UTILIZATION	0.2	2nd Decile
Average time to destination	1.4 hours	
Number of Destinations	42	
Number of breaks per 100 miles	1.8	

FIG. 11C

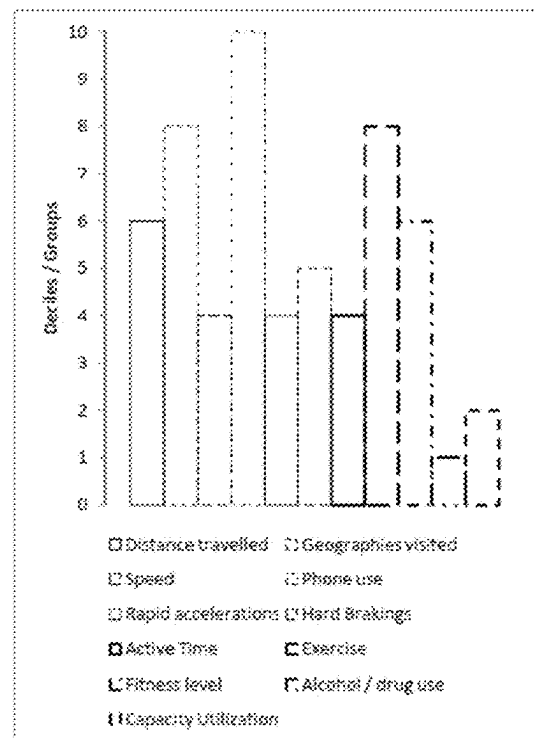


FIG. 11D

Vehicle Behavior		
	CUSTOMER	CLASSIFICATION GROUP
TOTAL VEHICLE BEHAVIOR	30	3rd Decile
Vehicle Identification number (VIN)	XXXXXXXXXXXX4352	
Vehicle make	Toyota	Group 4
Vehicle model	Camery	Group 22
Vehicle model year	2015	Group 12
Safety Features		Group 7
Passenger / cargo capacity	5	Group 5
Vehicle use	Commute	Group 2
Modifications	None	Group 1
Distance driven	47,000	Group 8
City miles	16,000	Group 1
Highway miles	31,000	Group 6
Maintainance	NA	Group 44
Manufacturerure's recall	3	Group 4
Number of owners	1	Group 1

FIG. 11E

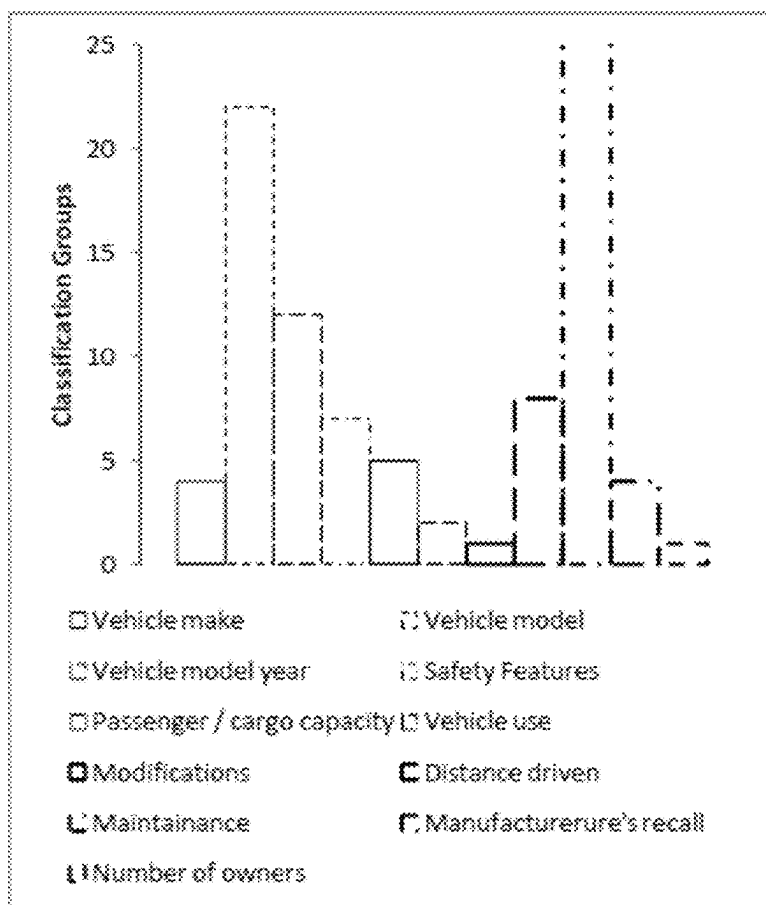


FIG. 11F

<u>Geographic Characteristics</u>		
	CUSTOMER	CLASSIFICATION GROUP
GEOGRAPHY TOTAL	50	5th Decile
Number of geographies	9	8th Decile
Demographic mix	E8	Group 4
Crime rate	H3	Group 12
Avg speed limits		Group 5
Roads <35mph	45%	
Roads with Speed 35-50mph	20%	
Roads with Speed 50+mph	35%	
Annual precipitation		
Rain	40 inches / year	Group 11
Snow	18 inches / year	Group 7
Average Temperature	58 °F	Group 15
Temperature < 35 °F	38 days	
Temperature 35-50 °F	125 days	
Temperature 50-80 °F	95 days	
Temperature > 80 °F	107 days	
Air Pollution levels	W3	9th Decile
Average elevation	16 feet	Group 2
Distance from coast / water	8,000 meters	Group 16
Health hazards	D5	Group 14

FIG. 11G

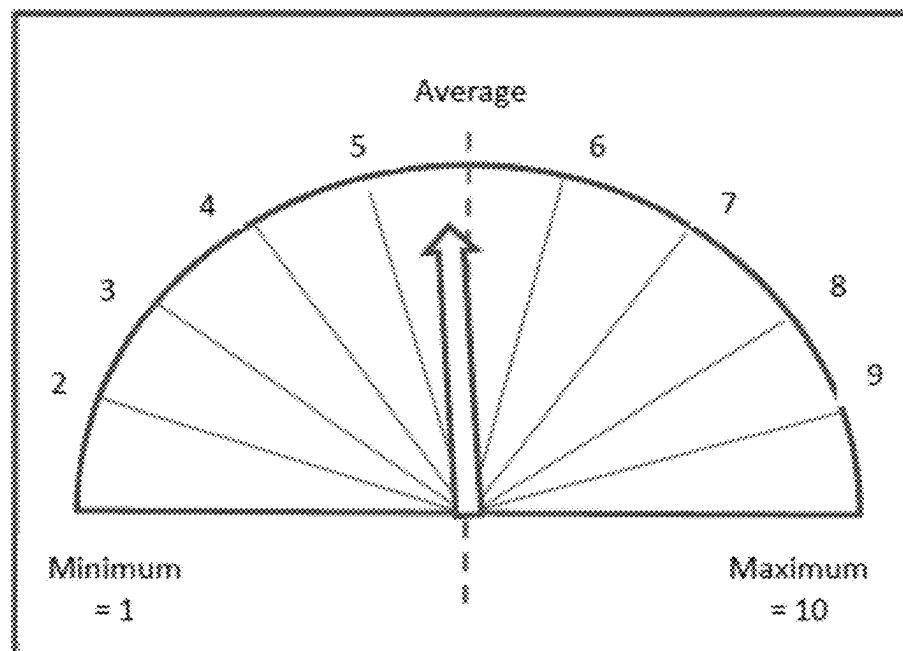


FIG. 11H

<u>Environmental Characteristics</u>			
	CLASSIFICATION		
	CUSTOMER	GROUP	
ENVIRONMENT TOTAL	35	3th Decile	
Number of geographies	9	8th Decile	
Light conditions	A6	Group 7	
Road work / accidents/ hazards	D2	Group 2	
Average precipitation			
Rain	2 inches	Group 8	
Snow	0 inches	Group 2	
Temperature conditions	66 °F	Group 15	
Road types	F2	Group 5	

FIG. 11I

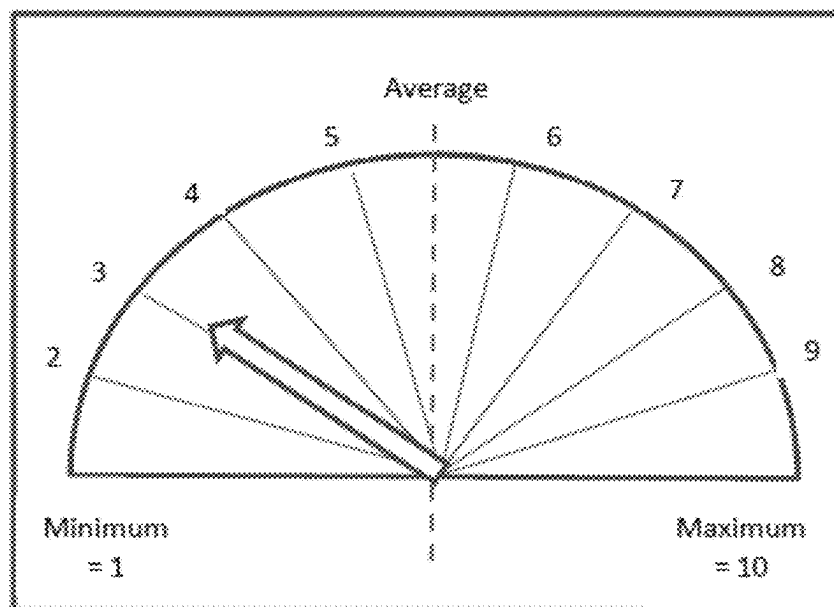


FIG. 11J

REPORT PERIOD

Start date

May 1 2015

End date

May 31 2015

CUSTOMER NUMBER

ABC123456789DM

EXERCISE	CLASSIFICATION	
	CUSTOMER	GROUP
Average active time	48 minutes	6th Decile
Days goal achieved	16	8th Decile
Average rest / sleep	7 hours	9th Decile
Body Temperature	98 °F	9th Decile
Blood Pressure	122/76	7th Decile
Heart rate	71	8th Decile
Blood glucose level	126	6th Decile
Blood alcohol level	0.002	4th Decile
Oxygen levels	97%	8th Decile

FIG. 11K

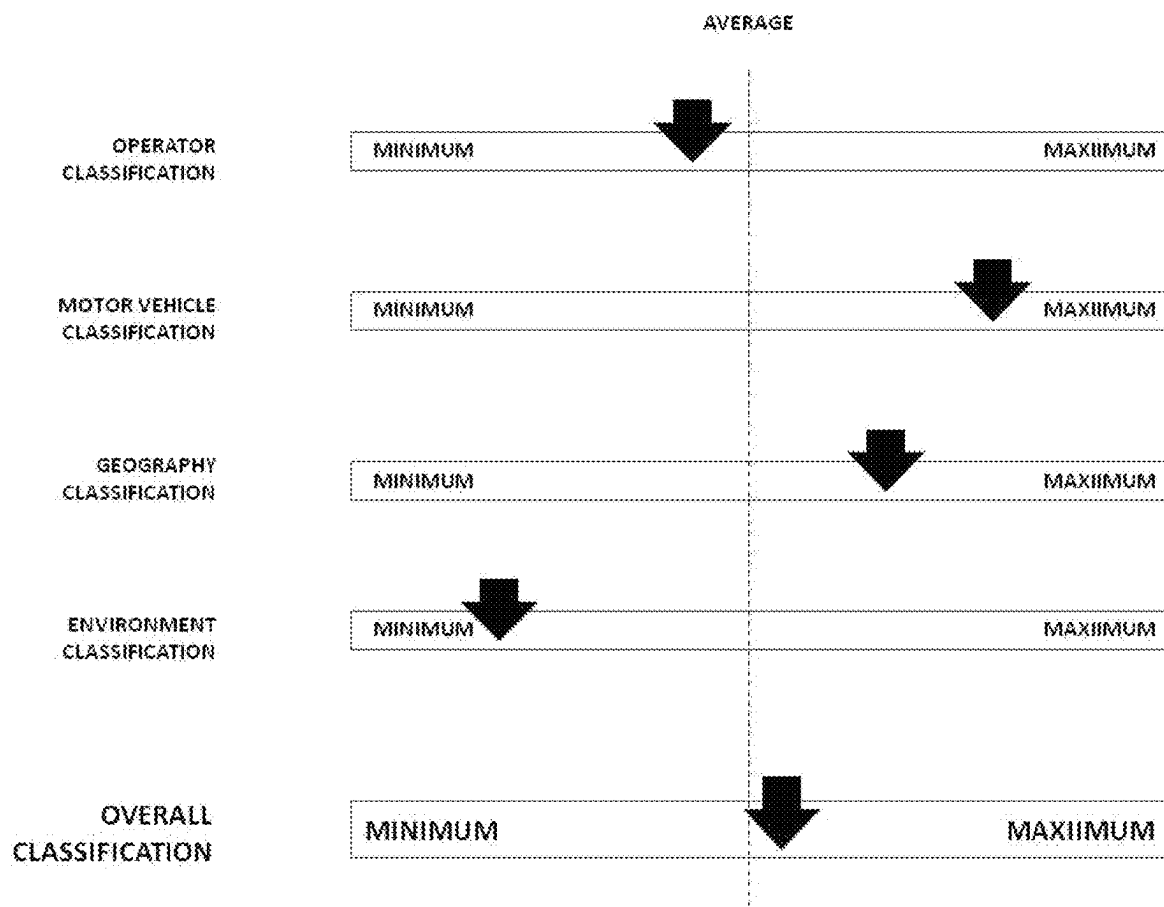


FIG. 11L

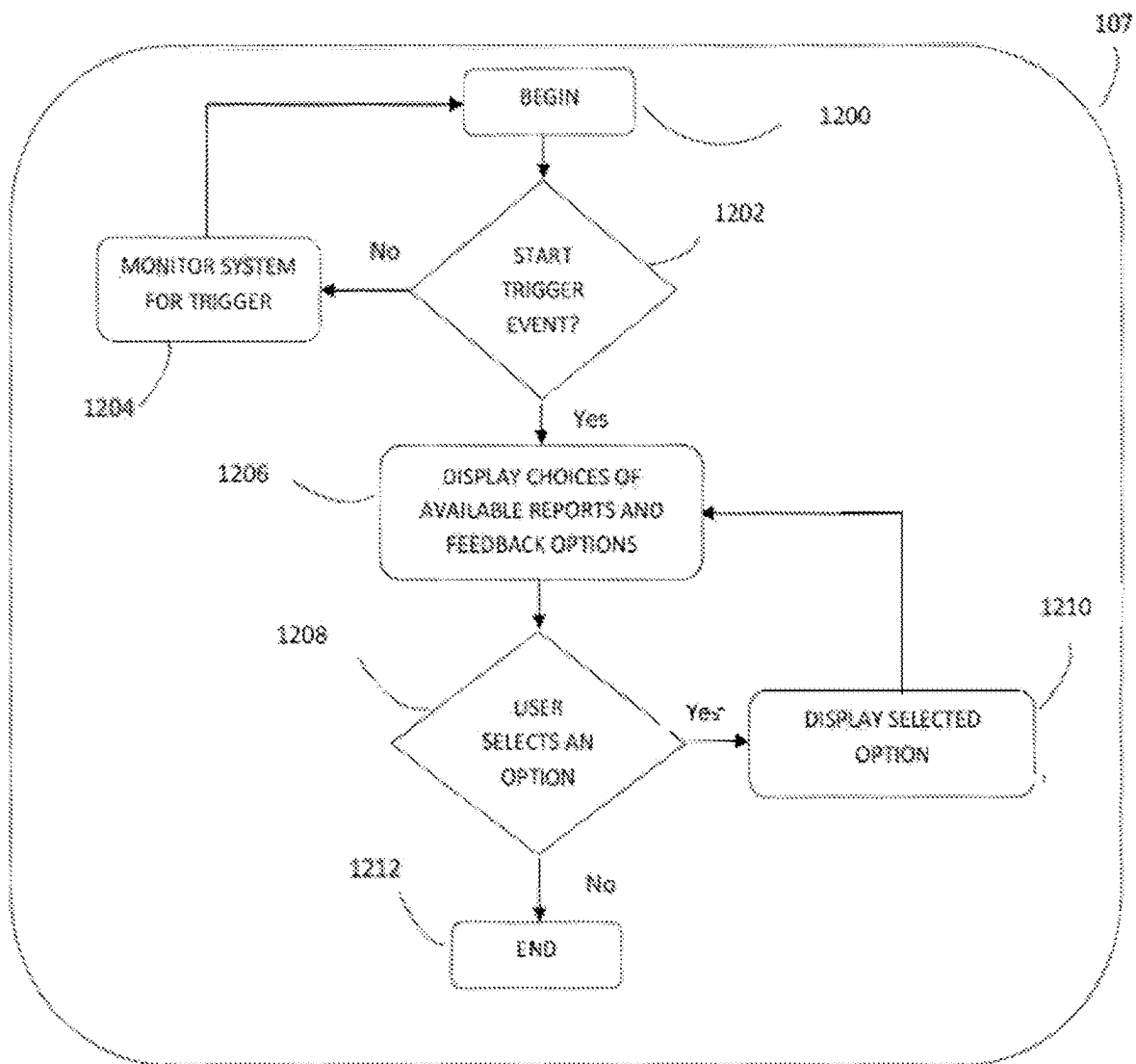


FIG. 12

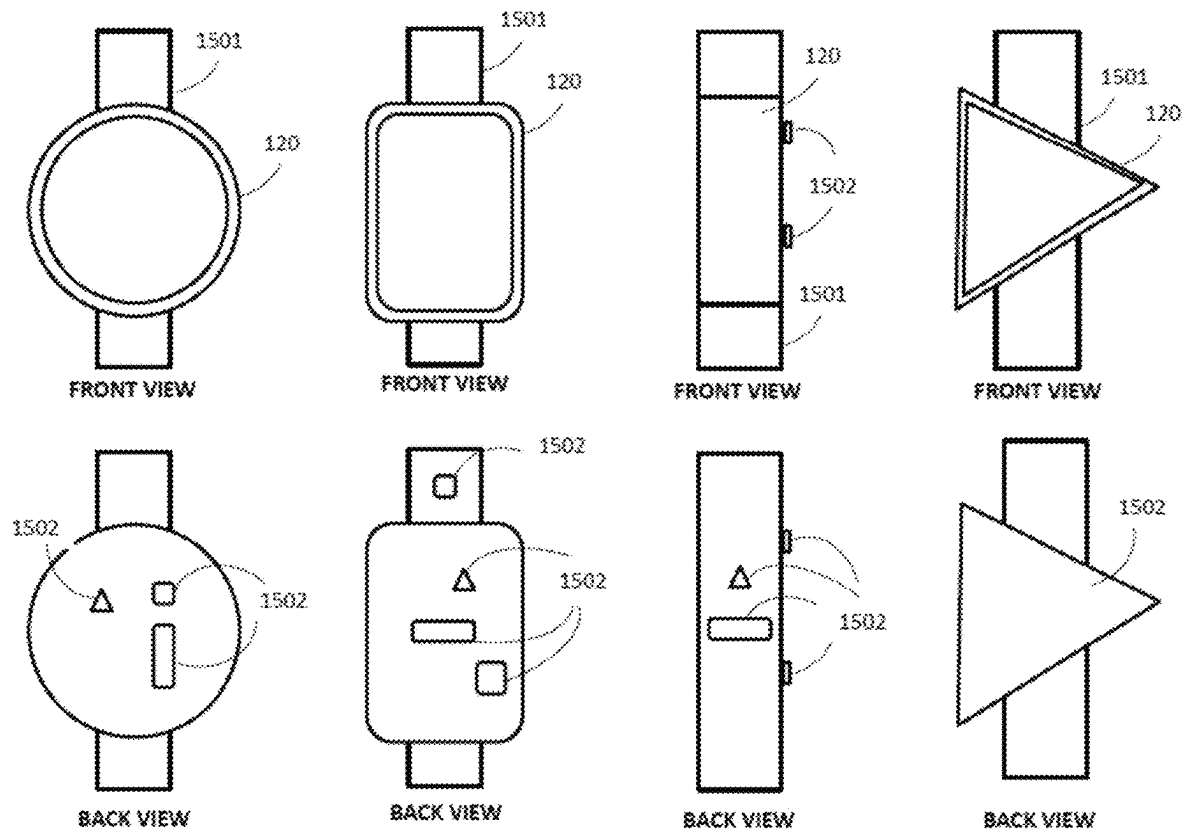


FIG. 13

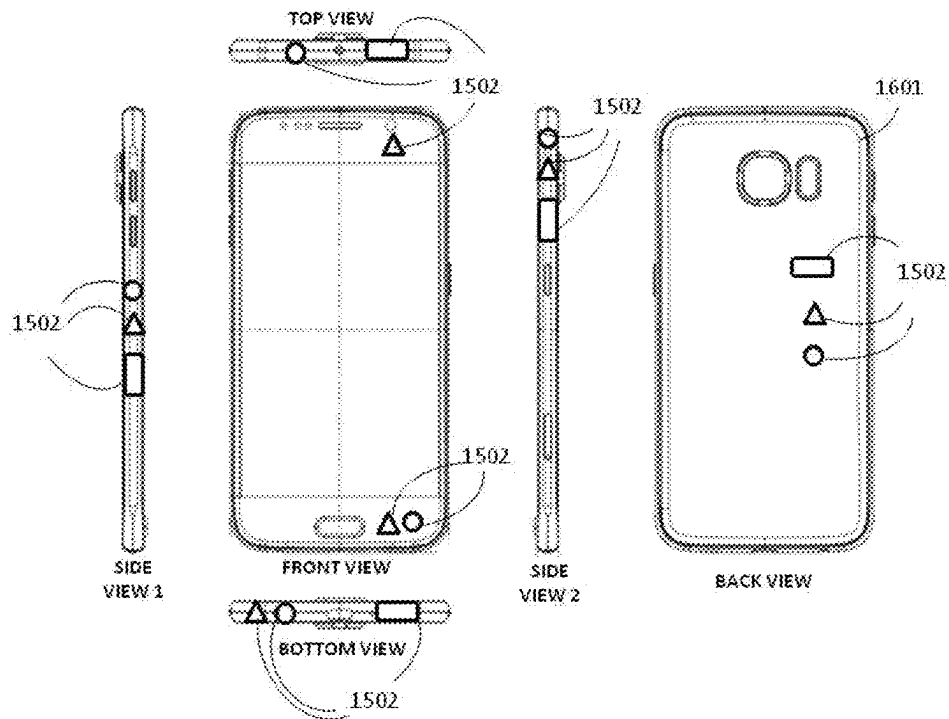


FIG. 14A

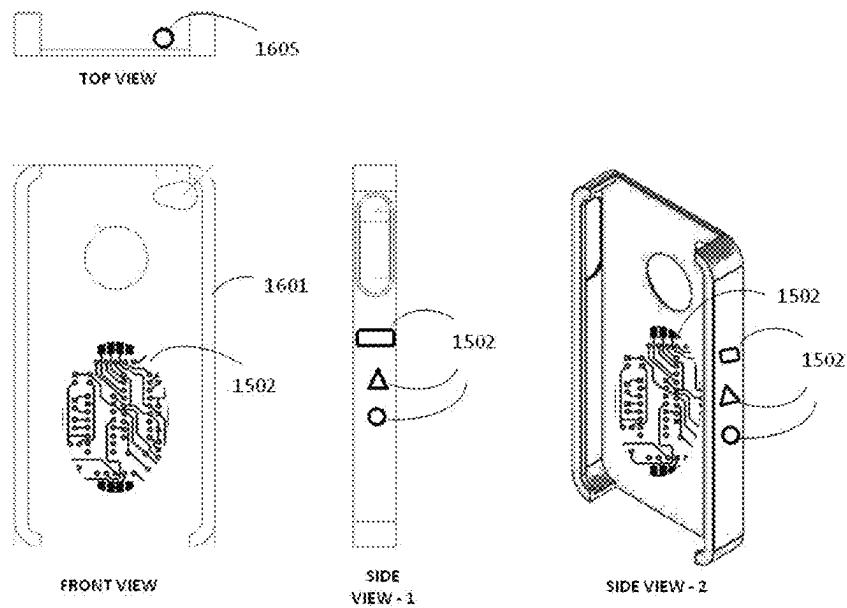


FIG. 14B

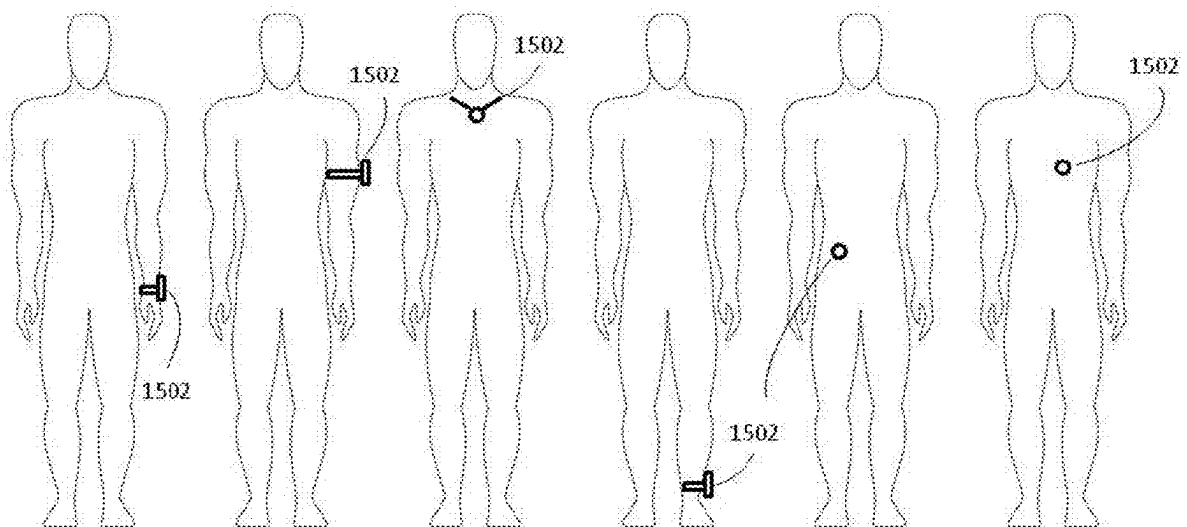
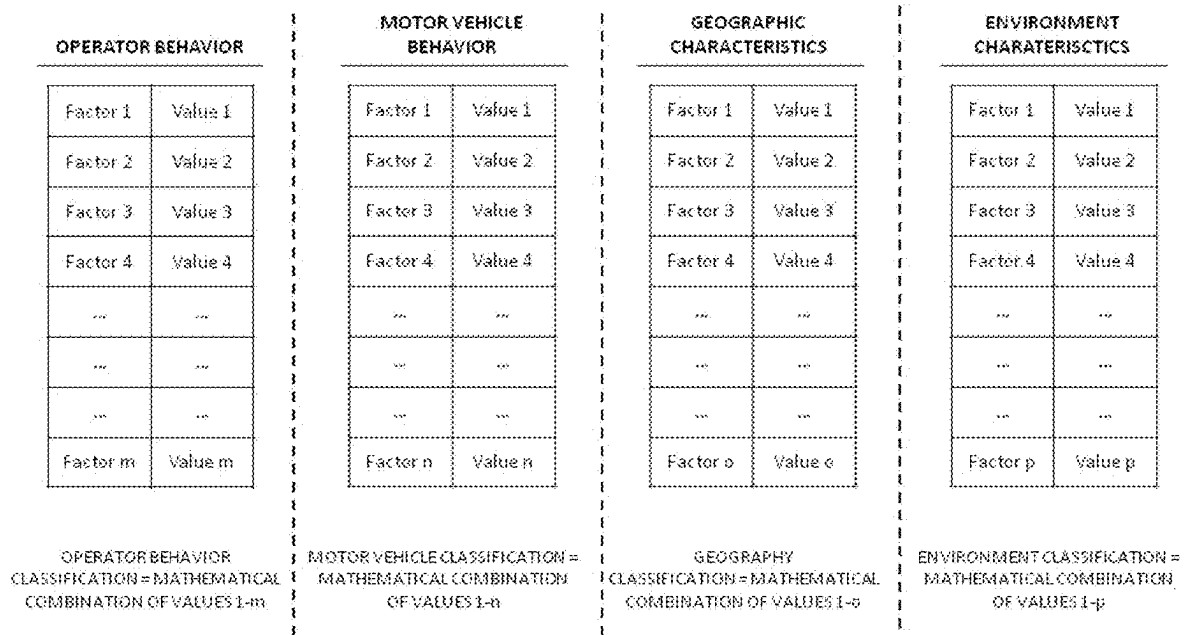


FIG. 15



COMBINED CLASSIFICATION = MATHEMATICAL COMBINATIONS OF OPERATOR BEHAVIOR, MOTOR VEHICLE BEHAVIOR, GEOGRAPHY CLASSIFICATION AND ENVIRONMENT CLASSIFICATION

FIG. 16

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SYSTEM FOR MONITORING AND CLASSIFYING VEHICLE OPERATOR BEHAVIOR

CROSS REFERENCE TO RELATED APPLICATIONS

The instant application is a continuation of and claims priority to U.S. patent application Ser. No. 16/288,884, filed on Feb. 28, 2019, which is a continuation of and claims priority to U.S. patent application Ser. No. 14/945,904, filed on Nov. 19, 2015, all of which are incorporated in their entirety by reference herein.

BACKGROUND OF THE INVENTION

The described invention relates in general to systems and methods for monitoring, evaluating and classifying vehicle operator behavior, vehicle behavior, details of geography, details of environment or combinations thereof and more specifically to a system for monitoring, evaluating and classifying vehicle operator behavior, motor vehicle behavior, details of geography, details of environment or combinations thereof using a mobile device such as a cell phone, smartphone or the like; a personal tracking device or a wearable electronic device such as smartwatch, fitness tracker or the like; and classifying the behaviors and details into groups based on quantitative patterns, relationships and interrelationships.

Conventional methods of monitoring the behavior of an operator typically require installation of one or more monitoring devices in the vehicle. Such devices may include an onboard computer and multiple sensors or data readers that plug into an on-board diagnostic port that collect vehicle operation data such as, for example, speed, location, and time of day. As such, these monitoring systems are not designed to be portable and cannot be easily transferred from one vehicle to another. Furthermore, incorrect installation may interfere with the normal and safe operation of the vehicle. Additionally, these monitoring systems may limit the monitoring and recording of data to the operation/function of the motor vehicle itself and may not monitor and record information regarding the operator's own behavior (such as texting or phone usage while driving), which may cause distraction and diminish the operator's ability to safely operate the motor vehicle. Furthermore, the systems are not designed to monitor the operator when the operator is not operating the vehicle or do not take into account lifestyle or health factors (such as sleep deprivation, shaky hands or blood alcohol level) that may further diminish operator's ability to safely operate a motor vehicle. Furthermore, the systems are not designed to collect vehicle location or environment information when it is not being operated by the operator, limiting the comprehensiveness of information collected and are thus not able to predict the likelihood of damage due to events like vandalism, theft or hail-storm, among others. Furthermore, current systems to monitor operator behavior lack efficient mechanism to classify all combinations of the operator behavior, the vehicle behavior, the geographic details and the environment details due to the extremely high computational power required and thus, either do not consider the geography and the environment where the vehicle is operated or consider only specific characteristics of the geographical and the environmental conditions, thereby limiting comprehensiveness of the classification for various combinations of the operator behavior, the vehicle behavior, the geographic details and the envi-

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ronment details. Therefore, there is an ongoing need for a system for monitoring and classifying vehicle operator behavior that is portable; does not require installation in the vehicle; includes information when the operator is operating the vehicle; includes information when the operator is not operating the vehicle; and provides an efficient mechanism to include comprehensive combinations of the operator behavior, the vehicle behavior, the geographic details, the environment details and combinations thereof.

SUMMARY OF THE INVENTION

The following provides a summary of certain exemplary embodiments of the present invention. This summary is not an extensive overview and is not intended to identify key or critical aspects or elements of the present invention or to delineate its scope.

In accordance with one aspect of the present invention, a system for determining qualification for insurance, cost of insurance or combinations thereof for a motor vehicle operator is provided. The system includes at least one personal monitoring device such as, for example, a smartwatch, a fitness tracker or other similar wearable portable electronic device; wherein the personal monitoring device includes one or more mechanisms for wearing the personal monitoring device by the operator and/or attaching the personal monitoring device to the operator such that the personal monitoring device is worn by the operator and/or attached to the operator when the operator is operating the motor vehicle and when the operating is not operating the motor vehicle, e.g., independently from the motor vehicle; wherein the personal monitoring device further includes hardware including a local memory, a battery, and sensors; wherein the sensors include a health and fitness sensor, an accelerometer, and a gyroscope; and the personal monitoring device provides functions including a Global Positioning System (GPS) function and a communication function; wherein the personal monitoring device is operative to monitor and communicate information about an operator of the motor vehicle, including at least driving data that is collected when the operator is operating the motor vehicle; wherein the driving data is solely collected by the personal monitoring device without receiving such driving data from vehicle sensors, calculated or derived based on data solely collected by the personal monitoring device without receiving such data from vehicle sensors, or combinations thereof; and wherein additional information including location is collected by the personal monitoring device when an operator is not operating the motor vehicle; wherein at least a portion of the information collected when the operator is not operating the motor vehicle is used to determine additional risk; wherein the system further includes at least one application software resident on the personal monitoring device; wherein the at least one application software interfaces with the personal monitoring device hardware and functions to monitor and communicate information used for determining qualification for insurance, cost of insurance or combinations thereof; wherein the at least one application software is activated and deactivated to perform at least one of the monitoring and communicating based on predetermined trigger events; and wherein the application software comprises a setup subsystem that is at least initiated when the user starts using the system; wherein the setup subsystem is operative to determine at least one trigger event used to activate or deactivate the application software; wherein the system further includes: at least one central control system that is operative to collect, process and communicate at least

a portion of the information collected and/or derived by the personal monitoring device used for determining qualification for insurance, cost of insurance or combinations thereof; at least one process to identify data issues in the information collected by the personal monitoring device used for determining qualification for insurance, cost of insurance or combinations thereof; wherein the data issues identified comprise presence of insufficient data, duplicate data, and data gaps; and at least one algorithm, wherein the algorithm is operative to determine qualification for insurance, cost of insurance or combinations thereof for the operator based on at least a portion of the information collected by the personal monitoring device.

In accordance with another aspect of the present invention, an alternate system for determining qualification for insurance, cost of insurance or combinations thereof for a motor vehicle operator is provided. The system includes at least one mobile device, such as, for example, a smartphone, a cellular phone, personal digital assistant or the like; wherein the mobile device is a portable device that is disposed in the motor vehicle but is not installed in the vehicle being monitored; at least one personal monitoring device, wherein the personal monitoring device includes one or more mechanisms for wearing the personal monitoring device and/or attaching the personal monitoring device to the operator such that the personal monitoring device is worn by the operator and/or attached to the operator when the operator is operating the motor vehicle and when the operator is not operating the motor vehicle, e.g., independently from the motor vehicle; wherein the personal monitoring device further comprises at least one health and fitness sensor, wherein the mobile device and the personal monitoring device are not installed in the motor vehicle being monitored and do not interface with the vehicle to collect information used for determining qualification for insurance, cost of insurance or combinations thereof; wherein the mobile device, the personal monitoring device or the combinations thereof comprises hardware including a local memory, a battery, and sensors; wherein the mobile device, the personal monitoring device or combinations thereof further provides functions including a Global Positioning System (GPS) function and a communication function; wherein the mobile device and the personal monitoring device are operative to at least collect and communicate information when the operator is operating the motor vehicle and when the operator is not operating the motor vehicle to determine qualification for insurance, cost of insurance or combinations thereof; wherein the information collected when the operator is operating the motor vehicle includes driving data; wherein the driving data is collected solely by the mobile device and the personal monitoring device without receiving such driving data from vehicle sensors; calculated or derived based on data solely collected by the mobile device and the personal monitoring device without receiving such data from vehicle sensors; or combinations thereof; and wherein the information collected when the operator is not operating the motor vehicle includes location; wherein the information collected when the operator is not operating the motor vehicle is used to identify additional information used for determining qualification for insurance, cost of insurance or combinations thereof; wherein the system further includes: at least one application software that resides on the mobile device and personal monitoring device; wherein the application software comprises a setup subsystem that is at least initiated when the user starts using the system; wherein the setup subsystem is operative to determine at least one trigger event used to activate or

deactivate the application software; at least one central control system that is operative to collect, process and communicate at least a portion of information collected and/or derived by the mobile device and the personal monitoring device used for determining qualification for insurance, cost of insurance or combinations thereof; and at least one process to identify data issues in the information collected by the mobile device, the personal monitoring device or combinations thereof including the presence of duplicate data and data gaps; and at least one algorithm, wherein the algorithm is operative to determine qualification for insurance, cost of insurance or combinations thereof based on at least a portion of information collected by the mobile device and the personal monitoring device.

In accordance with another aspect of the present invention, an alternate system for determining qualification for insurance, cost of insurance or combinations thereof for a motor vehicle operator is provided. The system includes at least one mobile device, wherein the mobile device is not installed within the motor vehicle; wherein the mobile device comprises hardware including a local memory, a battery, and sensors; wherein the sensors include an accelerometer and a gyroscope; wherein the mobile device further provides functions including a Global Positioning System (GPS) function and a communication function; wherein the mobile device is operative to at least collect and communicate information when the operator is operating the motor vehicle and when the operator is not operating the motor vehicle to determine qualification for insurance, cost of insurance or combinations thereof; wherein the information collected when the operator is operating the motor vehicle includes driving data; wherein the driving data is collected solely by the mobile device without receiving such driving data from vehicle sensors, calculated or derived based on data solely collected by the mobile device without receiving such data from vehicle sensors, or combinations thereof; and wherein the information collected when the operator is not operating the motor vehicle includes location; wherein at least a portion of the information collected when the operator is not operating the motor vehicle is used for determining qualification for insurance, cost of insurance or combinations thereof; wherein the system further includes at least one application software resident on the mobile device, wherein the application software is activated and deactivated to perform at least one of the said information collection and communication based on certain predetermined trigger events; and wherein the application software comprises a setup subsystem that is at least initiated when the user starts using the system; wherein the setup subsystem is operative to determine at least one trigger event used to activate or deactivate the application software; and wherein the system further includes: at least one central control system that is operative to receive, process and communicate at least a portion of the information collected and/or derived by the mobile device for determining qualification for insurance, cost of insurance or combinations thereof; and at least one process to identify data issues in the information collected by the mobile device including the presence of duplicate data and data gaps; and at least one algorithm, wherein the algorithm is operative to determine qualification for insurance, cost of insurance or combinations thereof based on at least a portion of the information collected by the mobile device when the operator is operating the motor vehicle and when the operator is not operating the motor vehicle.

Additional features and aspects of the present invention will become apparent to those of ordinary skill in the art upon reading and understanding the following detailed

description of the exemplary embodiments. As will be appreciated by the skilled artisan, further embodiments of the invention are possible without departing from the scope and spirit of the invention. Accordingly, the drawings and associated descriptions are to be regarded as illustrative and not restrictive in nature.

BRIEF DESCRIPTION OF THE DRAWINGS

The accompanying drawings, which are incorporated into and form a part of the specification, schematically illustrate one or more exemplary embodiments of the invention and, together with the general description given above and detailed description given below, serve to explain the principles of the invention, and wherein:

FIG. 1A is a block diagram of an exemplary embodiment of a system for monitoring and classifying vehicle operator behavior of the present invention illustrating the components and sub systems thereof;

FIG. 1B is a block diagram of an exemplary embodiment of a system for monitoring and classifying vehicle operator behavior of the present invention illustrating the components of the system that are resident on the handheld device, the components that are resident on the central control system and the means by which these components communicate internally and externally;

FIG. 1C is a block diagram illustrating that the central control system of the present invention may be in simultaneous communication with a plurality of handheld devices;

FIG. 1D is a block diagram of an exemplary embodiment of the present invention illustrating the components of system that are resident on the personal monitoring device; additional components resident on the handheld device; central control system; and that these components communicate internally and externally;

FIG. 1E is a block diagram of an exemplary embodiment of the present invention illustrating the information and components resident on the central control system; wherein a central control system may be in simultaneous communication with plurality of handheld devices, personal monitoring devices, or combinations thereof; wherein some of the information and components resident on a central control system may be updated from a plurality of external information sources;

FIG. 2 is a flow chart illustrating the operation of set-up subsystem of a system for monitoring and classifying vehicle operator behavior of the present invention;

FIG. 3A is a flow chart illustrating the operation of administrative control subsystem of a system for monitoring and classifying vehicle operator behavior of the present invention with regard to data collection;

FIG. 3B is a flow chart illustrating the operation of administrative control subsystem of a system for monitoring and classifying vehicle operator behavior of the present invention with regard to data transfer;

FIG. 3C is a flow chart illustrating the operation of administrative control subsystem of a system for monitoring and classifying vehicle operator behavior of the present invention with regard to reporting and feedback;

FIG. 4 is a flow chart illustrating the operation of data collection subsystem of a system for monitoring and classifying vehicle operator behavior of the present invention;

FIG. 5 is a flow chart illustrating the operation of data transmission subsystem of a system for monitoring and classifying vehicle operator behavior of the present invention;

FIG. 6 is a table presenting examples of various trigger events used in the operation of a system for monitoring and classifying vehicle operator behavior of the present invention;

FIG. 7A depicts an embodiment of the present invention wherein the user interface includes a handheld device such as mobile telephone, smartphone, personal digital assistant or the like;

FIG. 7B depicts an embodiment of the present invention wherein the user interface includes a handheld device such as a mobile telephone and a computer;

FIG. 7C depicts an embodiment of the present invention wherein the user interface includes a computer;

FIG. 7D depicts an embodiment of the present invention wherein the user interface includes a personal monitoring device and a mobile device;

FIG. 7E depicts an embodiment of the present invention wherein the user interface includes a personal monitoring device, such as, a smartwatch, fitness tracker, portable health monitor or wearable electronic device;

FIG. 8A depicts first sample data transmission file in accordance with a system for monitoring and classifying vehicle operator behavior of the present invention;

FIG. 8B depicts second sample data transmission file in accordance with a system for monitoring and classifying vehicle operator behavior of the present invention;

FIG. 8C depicts third sample data transmission file in accordance with a system for monitoring and classifying vehicle operator behavior of the present invention;

FIG. 9 is a first example of a trip summary report in accordance with a system for monitoring and classifying vehicle operator behavior of the present invention;

FIG. 10 is a second example of a trip summary report in accordance with a system for monitoring and classifying vehicle operator behavior of the present invention;

FIGS. 11A-11B illustrate a first set of examples of a usage summary report in accordance with a system for monitoring and classifying vehicle operator behavior of the present invention;

FIGS. 11C-11D illustrate a second set of examples of a usage summary report in accordance with a system for monitoring and classifying vehicle operator behavior of the present invention;

FIGS. 11E-11F illustrate a third set of examples of a usage summary report in accordance with a system for monitoring and classifying vehicle operator behavior of the present invention;

FIGS. 11G-11H illustrate a fourth set of examples of a usage summary report in accordance with a system for monitoring and classifying vehicle operator behavior of the present invention;

FIGS. 11I-11J is a fifth set of examples of a usage summary report in accordance with a system for monitoring and classifying vehicle operator behavior of the present invention;

FIG. 11K is a sixth example of a usage summary report in accordance with a system for monitoring and classifying vehicle operator behavior of the present invention;

FIG. 11L is a seventh example of a usage summary report in accordance with a system for monitoring and classifying vehicle operator behavior of the present invention;

FIG. 12 is a flowchart illustrating the operation of a reporting and feedback subsystem of the present invention;

FIG. 13 depicts examples of various designs or embodiments of personal monitoring device illustrating the locations of sensors of the present invention;

FIG. 14A illustrates an embodiment of personal monitoring device design illustrating that a personal monitoring device may be directly attached to mobile handheld device;

FIG. 14B illustrate an embodiment of personal monitoring device design illustrating that a personal monitoring device may be attached to a mobile phone case and/or indirectly attached to a mobile handheld device;

FIG. 15 is an illustration presenting examples of various location where personal monitoring device may be worn or attached to an operator in the operation of a system for monitoring and classifying vehicle operator behavior of the present invention; and

FIG. 16 is an illustration of an example of classification, groupings and calculations in accordance with a system for monitoring and classifying vehicle operator behavior.

DETAILED DESCRIPTION OF THE INVENTION

Exemplary embodiments of the present invention are now described with reference to the Figures. Reference numerals are used throughout the detailed description to refer to the various elements and structures. In other instances, well-known structures and devices are shown in block diagram form for purposes of simplifying the description. Although the following detailed description contains many specifics for the purposes of illustration, a person of ordinary skill in the art will appreciate that many variations and alterations to the following details are within the scope of the invention. Accordingly, the following embodiments of the invention are set forth without any loss of generality to, and without imposing limitations upon, the claimed invention.

The present invention relates to a system for monitoring and classifying the behavior of motor vehicle operator, the behavior of motor vehicle, the details of geography where the motor vehicle is operated, the details of environment where the motor vehicle is operated, and combinations thereof. Relevant data is collected using a mobile device, mobile telephone; smart phone; personable digital assistant; personal or wearable monitoring device (such as smartwatch or fitness tracker); other portable electronic devices; or combinations thereof. Relevant data collection may occur when the vehicle operator is operating the motor vehicle as well as other times when the vehicle operator is not operating the motor vehicle (vehicle is parked); or combinations thereof. The information collected or derived is transmitted to a remote server where the data is recorded, further processed, characterized, compared, classified and communicated to the vehicle operator and/or to another user of the system. The remote server may include additional data from third party sources and data providers. One or more algorithms included in this invention may be used to analyze data gathered by the system and assign classification or rating (or other characterization) to the operator, the vehicle, the geography, the environment or combinations thereof. The classifications or rating may be calculated by grouping behaviors and/or characteristics of the operators, the vehicles, the geographies, the environments or combinations thereof based on quantitative patterns, relationships and interrelationships. Information collected and/or derived may be further used for developing predictive models, decision models and/or descriptive models. Data recorded and/or derived from this invention may be used for purposes such as, for example, determining employability and compensation rate; providing travel and entertainment services; determining the insurability and cost of motor vehicle insurance; determining insurability and cost of health insurance; deter-

mining insurability and cost of life insurance; dispatching emergency services; commercial fleet management; route optimization; education; law enforcement; and monitoring and supervising drivers. The present invention gathers useful data from existing mobile devices such as smartphones, personal digital assistants and wearable and personal monitoring devices and does not require hard-wiring a device into a vehicle or other installation procedures, the incorrect execution of which may affect vehicle function or distract the vehicle's operator. Furthermore, the present invention gathers useful information about the health and fitness of the operator; and geography and environment of the operator and vehicle, even when the operator is not operating a motor vehicle. Application software resident on the mobile device and personal monitoring device component of this invention may also be used to independently verify the date and time using a variety of external sources including the central processing system, the mobile device (for personal monitoring device), the personal monitoring device (for mobile device), one or more atomic clocks, cellular service providers, or by other means. With reference now to the Figures, one or more specific embodiments of this invention shall be described in greater detail.

The block diagram provided in FIG. 1A illustrates the components and subsystems of an exemplary embodiment of the present invention. Set-up subsystem 100 is used by the motor vehicle operator to install and configure monitoring system 10. Once the monitoring system 10 is initially configured using the set-up subsystem 100, the monitoring system 10 then operates within the guidance and control of administrative subsystem 101. The administrative subsystem 101 guides and controls the operations of data collection subsystem 102, data transmission subsystem 103, local database 104, data transmission file 105 and reporting and feedback subsystem 107. The data transmission subsystem 103 uses the data transmission file 105 to communicate with central control system 106. The reporting and feedback subsystem 107 may provide reports and feedback to a user of the monitoring system 10 either from the local database 104 or from the central control system 106, or from a combination of both. FIG. 6 is a table presenting examples of various trigger events used by the subsystems of the monitoring system 10. These trigger events may be used to start a subsystem, stop a subsystem or modify the behavior of a subsystem.

The block diagram provided in FIG. 1B illustrates some of the components of a system 10 that are resident on handheld mobile device 700, some of the components that are resident on central control system 106 and the means by which these components communicate internally and externally. Local monitoring system 112 receives, processes, and stores information from a variety of local application software, system components, sensors, and/or local hardware available to a mobile device including those shown in FIG. 1B and FIG. 1D (or other components). Mobile device 700 may also communicate with one or more satellites 116 and/or mobile network 118. A central control system 106 typically includes the components shown in FIG. 1B and FIG. 1E (or other components) and communicates with a handheld device 700 by a variety of communication means including one or more cellular networks 118, satellites 116, radio signals, audio signals, electromagnetic signals, thermoelectric signals, Bluetooth, infrared, and/or internet connection 110, among others.

The block diagram in FIG. 1C illustrates that a central processing system 106 of the present invention may be in

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simultaneous communication with a plurality of handheld devices **700** at any given time.

The block diagram in FIG. **1D** illustrates the components of an exemplary embodiment of the present invention with personal monitoring device **120** such as smartwatch, fitness tracker, blood alcohol monitor or other portable electronic devices popularly known as wearable electronics; components **122** on a personable monitoring device **120**; components **119** on mobile device **700** in addition to components already illustrated in FIG. **1B**; central processing system **106**; and that these components communicate internally and externally. Local monitoring system **121** on the personal monitoring device **120** receives, processes, and stores information from a variety of local application software, system components, and/or local hardware and sensors available to the personal monitoring device including those shown in FIG. **1D** (or other components). Personal monitoring device **120** may operate independently or in combination with a handheld mobile device **700** to collect, process, store and communicate information relevant to the behavior and/or characteristics of the operator, the motor vehicle, the geography, the environment or combinations thereof; Personal monitoring device **120** may communicate with a handheld device **700** and/or the central control system **106** by a variety of short range and long range communication means including wired or wireless means such as cellular networks **118**, satellites **116**, Bluetooth, infrared, radio signals, audio signals, electromagnetic signals, thermoelectric signals, cellular network and/or internet, among others.

The block diagram in FIG. **1E** illustrates some of the information and components resident on a central control system **106** of an exemplary embodiment of the present invention. FIG. **1E** illustrates that the central control system **106** may be in direct or indirect communication with plurality of handheld mobile devices **700**; plurality of personal monitoring devices **120**; plurality of personal monitoring devices **120** working in combination with mobile devices **700**; or combinations thereof. Furthermore, the information resident on the central control system **106** may be updated or synchronized with a number of external information sources **150** including census information; national and local weather information; mapping and geographical information; traffic information; operator background and driving history (from sources like self reporting, law enforcement, department of motor vehicle, insurance companies, employment records, credit bureaus, third party information provider or combinations thereof); motor vehicle history (from sources like department of motor vehicle, vehicle manufacturer, current and prior owners, vehicle repair and service facilities, law enforcement, third party provider or combinations thereof); health and medical records of the operator (from sources like self reporting, national or local databases, medical service providers, medical research facilities, third party providers or combinations thereof); other third party information providers or combinations thereof. Central processing system **106** may communicate with internal and external components and data-sources **150** using a variety of short range and long range communication means including wired or wireless means such as cellular networks **118**, satellites **116**, Bluetooth, infrared, radio signals, audio signals, electromagnetic signals, thermoelectric signals, cellular network and/or internet, among others.

The flow chart provided in FIG. **2** illustrates the operations of set-up subsystem **100**. Set-up subsystem **100** is initiated at step **200** when a user first starts using the monitoring system **10** and each time the user wishes to re-configure the monitoring system **10** thereafter. At step

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201, set-up subsystem **100** acquires information such as, for example, confirmation of software license for use of system **10**; date of license grant (initial or renewed); vehicle identification number; vehicle make, vehicle model, vehicle model year, operator driving history, operator health and medical information, operator lifestyle, operator demographic information, among others. The set-up subsystem **100** also acquires user preferences at step **201**, which may include auto/manual updates to system software, real-time versus batch data transmission, data transmission preferences, data sharing, display reports, colors and themes, among others. The set-up subsystem **100** may also acquire information about multiple devices connected to the system **10** such as, for example, mobile device, smartwatch, fitness tracker; primary data source, and user preferences regarding their interconnectivity, data sharing, frequency of updates and synchronization. The set-up subsystem **100** may also acquire user preference and permissions about data sharing and data collection in reference with other users of the system **10** such as, for example, current or potential employer, motor insurance companies, health insurance company, life insurance company, travel and entertainment company, law enforcement, among others. At step **202**, the set-up subsystem **100** derives and calculates a plurality of control triggers for all of the subsystems of monitoring system **10**. At step **203**, the set-up subsystem **100** initiates administrative control subsystem **203**.

The flow charts provided in FIGS. **3A-3C** illustrate the functionality and operation of administrative control subsystem **101**, which controls the operations of all the other subsystems in a monitoring system **10**. The administrative control subsystem **101** monitors the system **10** for a plurality of trigger events to start, stop or modify the behavior of the other subsystems. In FIG. **3A**, the administrative control subsystem **101** controls the operation of data collection subsystem **102** by beginning at step **30**, acquiring control triggers and user preferences from the set-up subsystem **100** at step **300** and monitoring the system **10** for trigger events at step **301**. Decision step **302** is based on the presence of a trigger, or the lack thereof, to initiate the data collection subsystem **102**. Lack of a trigger results in no action at step **303**, while the presence of a "start" trigger initiates the data collection subsystem **102** at step **305**. Decision step **307** is based on the presence of a trigger, or the lack thereof, to terminate the data collection subsystem **102**. Lack of a trigger results in continued running of the data collection subsystem **102** at step **308**, while the presence of a "stop" trigger terminates the data collection subsystem **102** at step **310**. Examples of trigger events used by the data collection sub-system **102** includes time of day, speed above predetermined speed, change in heart rate, presence of blood alcohol above predetermined level, change in location, force start and force stop among others. Examples of additional triggers are included in table in FIG. **6**. In FIG. **3B**, the administrative control subsystem **101** controls the operation of data transmission subsystem **103** by beginning at step **30**, acquiring control triggers and user preferences from the set-up subsystem **100** at step **300** and monitoring the system **10** for trigger events at step **301**. Decision step **304** is based on the presence of a trigger, or the lack thereof, to initiate the data transmission subsystem **103**. Lack of a trigger results in no action at step **303**, while the presence of a "start" trigger initiates the data collection subsystem **103** at step **306**. Decision step **309** is based on the presence of a trigger, or the lack thereof, to terminate the data transmission subsystem **103**. Lack of a trigger results in continued running of the data transmission subsystem **103** at step **308**, while the

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presence of a “stop” trigger terminates the data transmission subsystem 103 at step 311. Examples of trigger events used by the data transmission sub-system 103 includes time of day, presence of local data file, proximity of multiple components of the system 10, file size, force start and force stop among others. Examples of additional triggers are included in table in FIG. 6. In FIG. 3C, the administrative control subsystem 101 controls the operation of reporting and feedback subsystem 107 by beginning at step 30, acquiring control triggers and user preferences from the set-up subsystem 100 at step 300 and monitoring the system 10 for trigger events at step 301. Decision step 314 is based on the presence of a trigger, or the lack thereof, to initiate the data reporting and feedback subsystem 107. Lack of a trigger results in no action at step 303, while the presence of a “start” trigger initiates the data reporting and feedback subsystem 107 at step 316. Decision step 318 is based on the presence of a trigger, or the lack thereof, to terminate the data reporting and feedback subsystem 107. Lack of a trigger results in continued running of the data reporting and feedback subsystem 107 at step 308, while the presence of a “stop” trigger terminates the data reporting and feedback subsystem 107 at step 320. Examples of trigger events used by the reporting and feedback sub-system 107 includes time of day, achievement of predetermined goal, presence of data, message from central control event, distance from predetermined location, system alerts, force start and force stop among others. Examples of additional triggers are included in table in FIG. 6. Additional triggers used by the system 10 includes presence of valid license agreement, datalink between components of monitoring system 10, battery level, number of attempts, restarting of a component of the monitoring system 10, among others. The administrative control subsystem 101 or other subsystems would use these triggers to start, stop or modify the behavior of the components of the present invention.

The flow chart provided in FIG. 4 illustrates the operations of the data collection subsystem 102, which collects data based on predetermined trigger events. Operation of data collection subsystem 102 begins at step 400 and proceeds to decision step 402. If no “start” trigger event has occurred, the data collection subsystem 102 continues to monitor system 10 for trigger events at step 401. If a “start” trigger event has occurred at decision step 402, the data collection subsystem 102 begins to collect operational data at step 403. This data is then stored in local database 104, which typically resides on a mobile phone, personal monitoring device or similar devices, at step 404. This cycle will continue until a “stop” trigger event occurs at step 405 at which point the data collection subsystem 102 will cease collecting operational data and/or other data. An example of a trigger event would be detecting predetermined speed of motion. Another example of a trigger event would be detecting physical activity like walking, running etc. Another example of a trigger event would be a system alarm or predetermined time intervals. The data collection subsystem 102 would use the “start” trigger to start collecting operational data. The data collection subsystem 102 is operative to monitor and record data when the operator is driving motor vehicle as well as when the operator is not driving the motor vehicle. The data collection subsystem 102 is typically operative to monitor and record certain data such as, for example, date; time; current geographic location; speed of motion; direction of motion; use of cellular phone by the operator; drug and alcohol use by the operator; amount of physical activity; and health and fitness of the operator. The data may then be used to calculate and derive

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or determine other relevant factors such as, for example, rapid acceleration; rapid deceleration; change in direction; speed of change in direction; distance travelled; total travel time; amount of sleep or rest; heart rate above or below predetermined levels; blood alcohol level above predetermined levels; level of alertness; level of physical activity above or below predetermined physical activity level; response time; change in relevant factors over time; among others. The data collection subsystem 102 typically utilizes multiple sensors and services (e.g. one or more internal sensors and various application software) resident on or available to (i.e., external to) the mobile device 700 and/or the personal monitoring device 120, or combinations thereof, such as, for example, a compass; an accelerometer; a gyroscope, a global positioning system (GPS); an atomic clock; temperature and weather monitoring; cell phone tower triangulation; a heart rate monitor; electrocardiogram; thermometer; blood alcohol monitor; blood oxygen monitor; blood glucose monitor; and internet access and datalink.

FIG. 5 is a flow chart illustrating the operations of the data transmission subsystem 103, which monitors and manages transmission files 105, which are resident on the mobile device or on the personal monitoring device 120. The transmission files 105 may be created by the various subsystems in the monitoring system 10. Operations of the data transmission subsystem 103 begins at step 500 and proceeds to decision step 501. If a transmission file 105 is available at step 501, the data transmission subsystem 103 then checks for the presence of a datalink and/or internet access at step 502. If a datalink and/or internet access is confirmed, the data transmission subsystem 103 then checks for the availability of central control system 106 or handheld mobile device 700 at step 503 based on where the data needs to be transmitted. If the availability of the central control system 106 or the handheld mobile device 700 is confirmed, the data transmission subsystem 103 will transmit the local transmission files 105 to the central control system 106 or the handheld device 700 at step 504 and then receive a transmission file 505 from the central control system 106 or the handheld mobile device 700 at step 505. The data transmission subsystem 103 may use one or more encryption technologies while communicating with internal and/or external components of the system 10. If the transmission file 505 confirms the data transmission was successful at step 506, the data transmission subsystem 103 will then determine the need to keep the local transmission file 105 at step 507 or to free up local resources (i.e., memory) by deleting the local transmission file 105. The data transmission subsystem 103 may utilize one or more data transmission technologies such as, for example, wired and wireless connections, internet, Wi-Fi, Bluetooth, infrared, radio signals, sound waves, electromagnetic signals, thermoelectric signals, cellular networks 118, satellites 116, among others. In some embodiments, data transmission occurs in real-time and in other embodiments, the transmission files 105 are stored in the mobile device 700 or the personal monitoring device 120 for a period of time prior to transmission.

With reference again to FIG. 1A-1E, the central control system 106 is operative to compare collected, calculated and/or derived data to known or predetermined parameters and conditions. Observed, calculated and/or derived information may include information relevant to the behavior and/or characteristics of the vehicle operator, the vehicle, the geography, the environment or combinations thereof, such as, for example: time of day; presence of accident; road construction; full or partial road closures; snowfall; rainfall; fog; health and fitness level; medical condition; drug or

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alcohol use; among others. Predetermined parameters and conditions may be grouped based on quantitative patterns, relationships and interrelationships such as, of example: vehicles driving within predetermined miles per hour groups; vehicles with predetermined cargo capacity groups; amount of driving and non driving time; total amount of driving; number of geographies visited; operators with predetermined amount of sleep deprivation; operators with predetermined fitness level groups; operators with predetermined amount of drug or alcohol in body groups; operators using predetermined amount of mobile phone while driving group; operators taking predetermined amount of rest/breaks groups; geographies with predetermined speed limit groups, geographies by distance from predetermined location groups (e.g. ocean, warehouse, school); type of terrain groups (e.g., long straight roads, meandering mountain roads); altitude from sea level group; time of day groups; amount of snowfall in predetermined period groups; among others. Observed behavior may be enhanced by adding self reported and/or third party information such as, of example: operators in predetermined demographic groups; operators in predetermined age groups; operators within predetermined credit score groups; operators with predetermined driving history groups (e.g. accidents, traffic violations, experience driving predetermined cargo/passenger or lack thereof); operators with predetermined medical condition groups; number of passengers or amount of cargo in the vehicle groups; actuarial information relevant to motor vehicle insurance; actuarial information relevant to health and medical insurance; actuarial information relevant to life insurance; life expectancy; mortality rates; among others. One or more algorithms within the central control system 106 may classify the operator, the vehicle, the geography, the environment or combinations thereof, differently at different times based on available information such as, for example: placing an operator operating a motor vehicle at a posted speed limit under one group or classification when no road construction is present and placing the same operator under another group or classification when road construction is present; placing an operator operating a motor vehicle at posted speed limit under one group or classification when there is no snowfall and placing the same operator under another group or classification when there is snowfall in excess of selected amount; among others. The behavior and/or characteristics may be placed in appropriate reference group for further processing, predictive modeling, decision modeling, descriptive modeling or combinations thereof.

Classifications and groupings may be designed for specific purpose or multiple purposes by selecting and grouping information relevant to the purpose from the information collected (observed, self reported, from third parties or combinations thereof) and/or derived. A user of the system 10 may be able to configure the system 10 for one or more specific purposes. In some embodiments the user may change the configurations and/or specific purposes multiple times while in other embodiments the configurations and/or specific purposes may not be changed after the initial selection. Further, some embodiments may allow a user to configure the system 10 to use for more than single purpose at the same time. An example of classifications and groupings, with regard to specific purpose such as for example vehicle fleet management, may include: operator availability; distance/time from a predetermined location groups; available capacity groups (e.g. ability to pick up passengers or cargo for transportation); type of goods that may be transported groups (e.g. liquids, dry, refrigerated, medical); experience with type of goods/passengers to be transported;

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desired time of delivery/pickup groups; and the amount of driving time and non-driving time recorded groups; among others. Another example of classifications and groupings, with regard to specific purpose, such as for example health and medical insurance, may include: medical history groups; family medical history groups; type of allergies group; gender; ethnicity; distance from medical facility offering predetermined services; amount of driving; type of driving (e.g. high/low risk driving, transport hazardous material, amount of cellular phone use while driving, etc.); air pollution levels measured in geographies visited; diseases and infections present in geographies visited; level of physical activity; lifestyle (e.g. amount of sleep, amount of alcohol use, etc.); actuarial information relevant to health and medical insurance; life expectancy; mortality rates; among others.

One or more algorithms may also provide classifications, ranking and rating based on relative classifications of the operators, the vehicles, the geographies, the environment or combinations thereof such as, for example: the average speed of operators using the system 10 could be determined for a predefined area at a predetermined time and a relative classification could be assigned to any operator deviating from the average speed; average ratio of driving time to resting time within predetermined time period (e.g. 24 hours, week among other) may be determined and a relative classification could be assigned to any operator deviating from the average amount of driving time to resting time ratio; average distance travelled within a predetermined time period (e.g., night, rush hour, day, month and year, among others) could be determined and a relative classification may be assigned to any operator deviating from the average distance travelled; average number of geographies visited within a predetermined time period (e.g. week, month, year among others) may be determined and relative classification may be assigned to any operator deviating from the average geographies visited; average cargo and/or passenger load for a predetermined class of vehicle could be predetermined and a relative classification may be assigned to any operator deviating from the average amount of cargo and/or passenger load for a class of vehicle; among others. Operator driving history such as, for example: a history of previous traffic incidents including accidents and traffic violations may also be used to determine or assign groups or classifications. Operator medical history such as, for example: history of allergies, medical conditions, previous illness, pregnancy and predispositions to certain illnesses or medical conditions based on family history may also be used to determine or assign groups or classifications.

One or more algorithms may further provide classifications, ranking and rating derived from information collected over time such as, for example: average behavior of an operator may be classified relative to behavior when geographic location is experiencing rain, snow or road construction; average driving speed may be classified under different light conditions; average driving and non-driving time in a geographic location may be classified relative to each other; average behavior during day, night and/or rush hour may be classified relative to each other; average behavior of an operator before, during and after predetermined actions or conditions (e.g., pregnancy; physical exercise; drug or alcohol use; texting or talking and driving; temporary or long term illness; commuting or vacationing among others) may be classified relative to each other; average behavior when the motor vehicle has been modified (e.g., for towing or carrying passenger and/or cargo; for farm use; for sporting

activity among others) may be classified relative to each other and average behavior when there are no such modifications, among others.

One or more algorithms may determine presence of duplicate information for the same activity such as, for example: mobile device and personal monitoring device both recording the amount of physical activity at the same time (e.g. running, walking) for an operator; and mobile device and personal monitoring device both recording speed of motion at the same time for an operator, among others. One or more algorithms may further determine presence of contradiction in the data collected through multiple components of the system **10** such as, of example: information from mobile device showing rest while information from personal mobile device showing physical activity for an operator; information from mobile phone and personal monitoring device showing different speed of motion for a vehicle; self reported and driving records showing different number of traffic violations; among others. One or more algorithms may further determine presence of information gaps or data errors, such as for example: information not collected when the battery resident of personal monitoring device is being charged; information gap when the operator is not carrying the mobile device; data errors due to low memory levels; data errors due to device operating system failure, among others. One or more algorithms may further determine presence of information biases, such as for example: information only collected or missing for certain time of day; sample data collected has greater/under representation of certain information; limited data collected for a demographic group; statistically insufficient data; among others. One or more algorithms may utilize various methodologies for data cleansing, sanitizing and enhancement to handle data issues such as, for example: suppress or eliminate appropriate data points; select primary data source; supplementary/complementary data; combine data; data sampling; clustering; or other statistical processes; among others.

One or more statistical and mathematical processes and/or algorithms may further provide grouping to the operator behavior, the vehicle behavior, the geography, the environment or any combinations thereof based on quantitative patterns, relationships and interrelationships as depicted in FIG. **16**. Use of statistical and mathematical processes to group and classify all operator behaviors, vehicle behaviors, geographical characteristics and environmental characteristics greatly reduces the computational power required compared to using every variation of operator behaviors, vehicle behaviors, geographical characteristics and environmental characteristics (which may be infinite) and thus, greatly improves the efficiency of the system **10** such as, for example, using distance travelled in groups of 1000 miles vs. using very single mile travelled independently; using number of steps walked/run per day in increments of 2000 steps vs. using every single step count independently; among others. Further, the use of statistical and mathematical processes to group and classify all information collected reduces computational power required to process the information as compared to using such methods on only a portion of information and not on others (e.g. only using statistical and mathematical processes to group and classify the operator behavior and the vehicle behavior but using every variation of geographical characteristics and environmental conditions). Further, the use of statistical and mathematical processes to group and classify all operator behaviors, vehicle behaviors, geographical characteristics and environmental characteristics produces more comprehensive results

compared to using such methods on only few combinations of operator behaviors, vehicle behaviors, geographical characteristics and environmental characteristics and using only specific characteristics of others i.e. not using all available information (e.g. using statistical and mathematical processes to group and classify the operator behavior and the vehicle behavior but only using specific characteristics of the geographical conditions and specific characteristics of the environmental conditions i.e. not using all available geographical and environmental conditions).

Statistical and mathematical processes may be further used to develop predictive models, decision models and/or descriptive models. Analytical techniques used to provide grouping, classifications, ranking and rating to the operators, the vehicles, the geographies, the environment or any combinations thereof may further include regressions techniques such as, for example: Linear regression model, Discrete choice model, Logistic model, Multinomial model, Probit model, Logit model, Time series model, Survival or duration model, Classification and regression trees, Multivariate models and Multivariate adaptive regression splines; machine learning techniques such as, for example: Neural networks, Multilayer Perceptron, Radial basis functions, Support vector machines, Naive Bayes, k-nearest neighbours and Geospatial predictive modeling; or combinations thereof.

Classification, ranking and rating may be used to determine employability in trade such as, for example: private and public transportation drivers; school bus drivers; police, ambulance, fire and/or emergency services providers; race car drivers; driving school instructors; hazardous material driver; trucking and cargo drivers; travelling sales personnel; insurance claims personnel; taxi drivers; and delivery service drivers; among others. Vehicle operators that meet predetermined criteria and/or rank above a predetermined threshold may qualify for employment while others that do not meet the predetermined criteria and/or rank below the threshold may not qualify for employment. Further, the classification, ranking and rating may be used to determine pay rate or compensation when vehicle operators are found to qualify for employment such as, for example: vehicle operators that are ranked or rated highest are paid and/or compensated more than vehicle operators that are ranked below them; average operator is paid predetermined pay rate and/or compensation while operators ranked higher and/or lower are paid or compensated on a relative basis; among others. Further, information collected or derived by system **10** may be used to periodically update the employability and compensation rate decision (e.g. insufficient information at the time of initial employment or compensation decision is updated after evaluation period; annual compensation is updated based on performance over predetermined period, etc.)

Classification, ranking and rating may be further designed for the purpose of determining perks, services, discount and/or surcharge in travel and tourism industry such as, for example: rental car companies using predetermined criteria to determine whether or not to rent to potential vehicle renter; rental car companies determining cost of rental and/or insurance; rental car companies determining rewards, discount or marketing offers; hotel and/or airline companies determining cost of their goods and/or services; hotel and/or airline companies determining rewards, discount or marketing offers; and travel services company determining rewards, discount or marketing offers, among others. Further, information collected or derived by the system **10** may be used to periodically update the pricing, reward or other

services and benefits (e.g. rental car companies offering lower price to operators driving in lower risk geographies or exhibiting lower risk driving behavior etc.)

Classification, ranking and rating may be further used to determine qualification and/or the cost of motor vehicle insurance. Vehicle operators that meet predetermined criteria and/or rank above a predetermined threshold may qualify for motor vehicle insurance while others that do not meet the predetermined criteria and/or rank below the predetermined threshold may not qualify for motor vehicle insurance offered by an insurance company (different motor vehicle insurance programs may use different thresholds). Further, the classification, ranking and rating may be used to determine the cost of insurance such as, for example: when vehicle operators are found to qualify for motor vehicle insurance by an insurance company, vehicle operators that are ranked or rated highest pay a lower cost for motor vehicle insurance compared to vehicle operators that are ranked below them; average drivers is charged predetermined cost of insurance while drivers ranked higher and/or lower are charged on a relative basis; among others. Further, information collected or derived by the system 10 may be used to periodically update the insurability and/or cost of motor vehicle insurance (e.g. motor vehicle insurance premium is updated periodically, e.g. at renewal, based on updated classification or ranking; insufficient information at the time of initial motor vehicle insurance sale is updated after evaluation period; driving behavior is used to determine fault after the operator is involved in an motor vehicle accident and motor vehicle insurance premium is adjusted based on predetermined criteria, etc.)

Classification, ranking and rating may be further used to determine qualification and/or the cost of medical and health insurance. System 10 users that meet predetermined criteria and/or rank above a predetermined threshold may qualify for health insurance while others that do not meet the predetermined criteria and/or rank below the predetermined threshold may not qualify for health insurance offered by an insurance company (different health insurance programs may use different thresholds). Further, the classification, ranking and rating may be further used to determine the cost of insurance by an insurance company, such as, for example: when a user of the system 10 is found to qualify for health insurance by an insurance company, system 10 users that are ranked or rated highest pay a lower premium for health insurance compared to system 10 users that are ranked below them; average user is charged predetermined cost of insurance while users ranked higher and/or lower are charged on a relative basis; among others. Further, information collected or derived by the system 10 may be used to periodically update the insurability and/or cost of health insurance (e.g. health insurance premium is updated periodically, e.g. annually, based on updated classification or ranking; insufficient information at the time of initial health insurance sale is updated after evaluation period and health insurance premium adjusted accordingly, etc.)

Classification, ranking and rating may be further used to manage fleet operations such as, for example: align vehicle operators and projects based on operator classification and history; determine best route to transport cargo or passengers between predetermined locations; determine new routes to transport cargo or passengers between predetermined locations when route determined earlier is no longer available; provide new cargo when already dispatched cargo is unable to reach predetermined location on time and/or with desired quality; to provide dispatch and/or passenger pickup services; determine best route to intercept criminal and/or

illegal activities; determine best way to provide logistical support to predetermined locations and/or facility; and determine best route to evacuate in case of emergency; among others.

Classification, ranking and rating may be further used to provide marketing services such as, for example: location based services (e.g. food, shopping, sports, entertainment, transportation, tourism and hotels among others); delivery services (e.g. food, groceries, shopping, medicines, cargo and personal services among others) qualification for motor vehicle insurance; cost of motor vehicle insurance; employment offers; compensation; fleet management; and educational among others.

Classification, ranking and rating may be further used to provide educational programs such as, for example: young driver education; driving license examination (e.g. operator behavior observed for extended period of time instead of one time driving exam; operators need to cross predetermined thresholds, operators are required to exhibit predetermined behavior over predetermined period of time, etc.); rehabilitation after injury and/or illness; driver safety education (e.g. exhibit safe driving behavior over predetermined period); food and nutrition education; commercial driving education to non-commercial drivers; operating specialized motor vehicles; and operating predetermined type of vehicles; among others. (e.g. vehicle operators at the beginning of the educational program is given initial classification; this classification is updated based on operator exhibiting predetermined behavior or meeting predetermined thresholds).

Classification, ranking and rating may be further used to provide emergency services such as, for example: dispatch ambulance when health and fitness of the vehicle operator indicates certain predetermined signs (e.g. cardiac arrest); dispatch fire, police and/or ambulance in case of accident; provide relief supplies when stranded in remote location; provide evacuation services after emergency; and search and/or find services among others.

Classification, ranking and rating may be further used for law enforcement such as, for example: determine and take action against traffic violators; determine and take action against court ordered geographic boundary and/or limitation violators; determine criminal and/or illegal trespassing; and determine when a known suspect and/or fugitive is located within jurisdiction; among others. Examples of law enforcement applications include: vehicle operator exceeding speed limit on a given trip is charged appropriate penalty after the trip even when no law enforcement office caught the operator speeding during the trip; vehicle operator operating the vehicle with blood alcohol level exceeding legal limit is charged appropriate penalty after the trip even when no law enforcement office caught the operator during the trip; fault for an accident is determined based on vehicle operator behavior, among others.

FIG. 7A depicts an embodiment of the present invention wherein the user interface with the central control system 106 includes a handheld device such as mobile telephone 700 having a visual display 701. FIG. 7B depicts an embodiment of the present invention wherein the user interface with the central control system 106 includes a handheld device such as mobile telephone 700, which is in communication with a computer 702 having a visual display 703. FIG. 7C depicts an embodiment of the present invention wherein the user interface with the central control system 106 includes a computer 702 having a visual display 703. FIG. 7D depicts an embodiment of the present invention wherein the user interfaces with the central control system 106 includes a personal monitoring device 120 and a mobile device 700.

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FIG. 7E depicts another embodiment of the present invention wherein the user interfaces with the central control system **106** includes a personal monitoring device **120**. Some or all of these embodiments may be used for the initial setup of the monitoring system **10**, viewing trip summaries and/or performance reports, upgrading software, and sending information to and receiving information from the central control system **106**, among other purposes.

FIG. 8A depicts a sample data transmission file that includes a plurality of records, each containing some or all of the data recorded and/or derived from the data recorded by the data collection sub system **102** including date, time, vehicle speed, speed limit, zip code, direction, use of mobile phone, latitude, longitude, step count, and active time during the day. FIG. 8B depicts another sample data transmission file that includes a plurality of records, each containing some or all of the data recorded and/or derived from the data recorded by the data collection sub-system **102** including, date, time, blood pressure, blood glucose level, body temperature, heart rate, and blood alcohol level. FIG. 8C depicts another sample data transmission file that includes a plurality of records, each containing some or all of the data recorded and/or derived from the data recorded by data collection sub-system **102** including, date, time, body temperature, awake time, active time, walking step count, running step count, stairs or vertical step count, calories burnt, distance travelled. FIG. 9 is a first example of a trip summary report recorded by the data collection subsystem **102** showing a starting point for a trip, the route followed, and ending point of the trip on a map. FIG. 10 is a second example of a trip summary report recorded by the data collection subsystem **102** showing the actual speed of the vehicle and the speed limit posted on the route followed. The graphical representation may also show periods of rapid acceleration, deceleration, or hard braking. The graphical representation may include date and time of the trip. FIGS. 11A-11L provide various examples of usage reports comparing the operator, the vehicle, the geography and the environment performance to others. These reports may include summaries of various behaviors and characteristics collected including calculated and/or derived values and classifications such as, for example: an individual operator's behavior may include information about operator's driving history including miles driven; driving time; average speed; number of rapid acceleration; number of hard braking; health and fitness level; active time; driving in sleep deprivation state; calculated group ranking or classification; graphical representations of classification; graphical representation of the operator performance compared to other operators or groups; use of cellular phone; and geographies visited; among others. Reports may classify, rate and/or group the driver behavior, the vehicle behavior, the geographical characteristic and the environmental conditions based on quantitative patterns, relationships and interrelationships. The reports may include classification information about groups such as for example: the group behavior; relative ranking of the group compared to other groups; optimal range; relative risk; and productivity; among others. Reports may also include classification information based on the specific configuration of the system **10** such as, for example: specific type of employment score and/or group and information relevant to employment score and/or group; travel and entertainment score and/or group and information relevant to travel and entertainment score and/or group; risk score and/or group for motor vehicle insurance and information relevant to motor vehicle insurance score and/or group; risk score and/or group for health insurance and

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information relevant to health insurance score and/or group; risk score and/or group for life insurance and information relevant to life insurance score and/or group; number of milestones achieved and/or need to achieve to get driving license; among others.

FIG. 12 is a flowchart illustrating the operation of the reporting and feedback subsystem **107**. The operation of reporting and feedback subsystem **107** begins at step **1200** and proceeds to decision step **1202**. If no "start" trigger event has occurred, the reporting and feedback subsystem **107** continues to monitor the system **10** for trigger events at step **1204**. If a "start" trigger event has occurred at decision step **1202**, the reporting and feedback subsystem **107** displays choice of available reports and feedback options at step **1206**. The user selects one or more options at decision step **1208**, which results in the selected option being displayed at step **1210** or the process ending at step **1212**.

FIG. 13 illustrates examples of various designs of the personal monitoring device **120** and location of various sensors **1502** (e.g. health, fitness and medical sensors, among others) on the personal monitoring device **120** of a vehicle operator monitoring and classification system of the present invention. The personal monitoring device **120** design includes watch, band and/or personal clothing item (e.g. hat, shirt, gloves, shoe among others). The personal monitoring device includes one or more mechanism for the operator to wear or attach the device to themselves including a strap, a band, Velcro, tape or equivalent **1501**. The personal monitoring device may be worn; attached directly or indirectly (e.g. over clothing item); kept in close proximity; or combinations thereof, to the operator at various locations as illustrated in FIG. 15 (among others). The sensors **1502** resident on the personal monitoring device may be arranged in various positions/locations including back, side, band or combinations thereof. The sensors **1502** may or may not be placed externally or placed inside a protective covering and thus may or may not be visible to operator as depicted in FIG. 13.

FIG. 14A illustrates an embodiment of the present invention that includes the personal monitoring device **120** and the resident sensors **1502** that may be temporarily and/or permanently attached to the handheld device mobile device **700** like a smartphone. The location of attachment includes front, back, sides, top, bottom, or combinations thereof. The personal monitoring device **120** may use one or more port and/or interface available on the handheld mobile device **700** to communicate and/or interface with the mobile device **700**. The personal monitoring device **120** may include its own battery and/or power source, use mobile device **700** to draw power or combinations thereof.

FIG. 14B illustrate another embodiment of the present invention illustrating that the personal monitoring device **120** may be attached to and/or designed as a case **1601** for the handheld mobile device **700** such as a smartphone. The electrical circuitry **1502** is housed inside the case's design. The personal monitoring device **120** includes interface **1605** that connects with the mobile device **700**. The personal monitoring device **120** may include its own battery and/or power source, use mobile device **700** to draw power or combinations thereof.

While the present invention has been illustrated by the description of exemplary embodiments thereof, and while the embodiments have been described in certain detail, it is not the intention of the Applicant to restrict or in any way limit the scope of the appended claims to such detail. Additional advantages and modifications will readily appear to those skilled in the art. Therefore, the invention in its

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broadier aspects is not limited to any of the specific details, representative devices and methods, and/or illustrative examples shown and described. Accordingly, departures may be made from such details without departing from the spirit or scope of the Applicant's general inventive concept.

What is claimed is:

1. A system for determining qualification for insurance, cost of insurance or combinations thereof for a motor vehicle operator, comprising:

(a) at least one personal monitoring device,

(i) wherein the personal monitoring device includes one or more mechanisms for wearing the personal monitoring device and/or attaching the personal monitoring device to the operator such that the personal monitoring device is worn by the operator and/or attached to the operator when the operator is operating the motor vehicle and when the operating is not operating the motor vehicle;

(ii) wherein the personal monitoring device is not installed in the motor vehicle when the operator is operating the motor vehicle;

(iii) wherein the personal monitoring device further comprises hardware including a local memory, a battery, sensors, and a Global Positioning System (GPS) component; wherein the sensors include a health and fitness sensor, an accelerometer, and a gyroscope; and the personal monitoring device provides functions including a GPS function and a communication function;

(iv) wherein the personal monitoring device is operative to monitor and communicate information about an operator of the motor vehicle, including at least driving data that is collected when the operator is operating the motor vehicle;

(v) wherein the driving data is solely collected by the personal monitoring device without receiving such driving data from vehicle sensors, calculated or derived based on data solely collected by the personal monitoring device without receiving such data from vehicle sensors, or combinations thereof; and

(vi) wherein additional information including location is collected by the personal monitoring device when an operator is not operating the motor vehicle; wherein at least a portion of the information collected when the operator is not operating the motor vehicle is used to determine additional risk;

(b) at least one application software resident on the personal monitoring device,

(i) wherein the application software interfaces with the personal monitoring device hardware and functions to monitor and communicate information used for determining qualification for insurance, cost of insurance or combinations thereof;

(ii) wherein the application software is activated and deactivated based on predetermined trigger events; and

(iii) wherein the application software comprises a setup subsystem that is at least initiated when the user starts using the system; wherein the setup subsystem is operative to determine at least one trigger event used to activate or deactivate the application software;

(c) at least one central control system that is operative to collect, process and communicate at least a portion of the information collected and/or derived by the per-

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sonal monitoring device used for determining qualification for insurance, cost of insurance or combinations thereof;

(d) at least one process to identify data issues in the information collected by the personal monitoring device;

wherein the information collected and processed for identifying data issues comprises driving data including date, time and speed collected by one or more components of the personal monitoring device including GPS, accelerometer, or combinations thereof;

wherein the data issues identified comprise:

presence of insufficient driving data collected by one or more components of the personal monitoring device; and

presence of duplicate driving data, gaps in driving data, or combinations thereof recorded by the personal monitoring device; and

(e) at least one algorithm, wherein the algorithm is operative to determine qualification for insurance, cost of insurance or combinations thereof for the operator based on at least a portion of the information collected by the personal monitoring device.

2. The system of claim 1, further comprising at least one mobile device,

(a) wherein the at least one mobile device is not installed in the motor vehicle being monitored;

(b) wherein the mobile device further comprises hardware including a gyroscope, local memory, and a battery, and further provides functions including a Global Positioning System (GPS) function and a communication function;

(c) wherein the mobile device further includes at least one application software resident on the mobile device; and

(d) wherein the mobile device is operative to monitor and communicate additional information used to determine qualification for insurance, cost of insurance or combinations thereof for the operator.

3. The system of claim 1, wherein the personal monitoring device is a smartwatch; a fitness tracker; a portable medical, health and fitness device; or combinations thereof.

4. The system of claim 2, wherein the mobile device is a smartphone.

5. The system of claim 1, wherein the insurance includes motor vehicle insurance.

6. The system of claim 1, wherein the insurance includes health insurance.

7. The system of claim 1, wherein the insurance includes life insurance.

8. The system of claim 1, wherein the system is further operative to collect at least one use of personal monitoring device, mobile device or combinations thereof when the operator is operating the motor vehicle.

9. The system of claim 1, wherein the gaps in driving data are due to failure of the personal monitoring device components to capture the information, failure of the operator to carry the personal monitoring device, or combinations thereof.

10. The system of claim 1, wherein the gaps in driving data are due to: information not collected when the battery of the personal monitoring device is being charged; an information gap when the operator is not carrying the personal monitoring device; data errors due to low memory levels; data errors due to operating system failure of the personal monitoring device; or a combination.

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11. A system for determining qualification for insurance, cost of insurance or combinations thereof for a motor vehicle operator, comprising:

- (a) at least one mobile device; wherein the mobile device is a portable device that is disposed in the motor vehicle but is not installed in the vehicle being monitored;
- (b) at least one personal monitoring device,
 - (i) wherein the personal monitoring device includes one or more mechanisms for wearing the personal monitoring device and/or attaching the personal monitoring device to the operator such that the personal monitoring device is worn by the operator and/or attached to the operator when the operator is operating the motor vehicle and when the operator is not operating the motor vehicle; and
 - (ii) wherein the personal monitoring device further comprises at least one health and fitness sensor;
- (c) wherein the mobile device and the personal monitoring device are not installed in the motor vehicle being monitored and do not interface with the vehicle to collect information used for determining qualification for insurance, cost of insurance or combinations thereof;
- (d) wherein the mobile device, the personal monitoring device or the combinations thereof comprises hardware including a local memory, a battery, a Global Positioning System (GPS) component, and sensors including an accelerometer; wherein the mobile device, the personal monitoring device or combinations thereof further provides functions including a GPS function and a communication function;
- (e) wherein the mobile device and the personal monitoring device are operative to at least collect and communicate information when the operator is operating the motor vehicle and when the operator is not operating the motor vehicle to determine qualification for insurance, cost of insurance or combinations thereof,
 - (i) wherein the information collected when the operator is operating the motor vehicle includes driving data; wherein the driving data is collected solely by the mobile device and the personal monitoring device without receiving such driving data from vehicle sensors; calculated or derived based on data solely collected by the mobile device and the personal monitoring device without receiving such data from vehicle sensors; or combinations thereof; and
 - (ii) wherein the information collected when the operator is not operating the motor vehicle includes location; wherein the information collected when the operator is not operating the motor vehicle is used to identify additional information used for determining qualification for insurance, cost of insurance or combinations thereof;
- (f) at least one application software that resides on the mobile device and personal monitoring device; wherein the application software comprises a setup subsystem that is at least initiated when the user starts using the system; wherein the setup subsystem is operative to determine at least one trigger event used to activate or deactivate the application software;
- (g) at least one central control system that is operative to collect, process and communicate at least a portion of information collected and/or derived by the mobile device and the personal monitoring device used for determining qualification for insurance, cost of insurance or combinations thereof;

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- (h) at least one process to identify data issues in the information collected by the mobile device, the personal monitoring device or combinations thereof; wherein the information collected and processed for identifying data issues comprises driving data including date, time and speed collected by one or more components of the mobile device and/or personal monitoring device including GPS, accelerometer, or combinations thereof;

wherein the data issues identified comprise:

- presence of insufficient driving data collected by one or more components of the mobile device and/or personal monitoring device; and
- presence of duplicate driving data, gaps in driving data, or combinations thereof recorded by the mobile device and/or personal monitoring device; and

- (i) at least one algorithm, wherein the algorithm is operative to determine qualification for insurance, cost of insurance or combinations thereof based at least on a portion of information collected by the mobile device and the personal monitoring device.

12. The system of claim 11, wherein the mobile device is a smartphone and the personal monitoring device is a smartwatch; a fitness tracker; a portable medical, health and fitness device; or combinations thereof.

13. The system of claim 11, wherein the system is further operative to collect the use of mobile device, personal mobile device or combinations thereof by the operator when driving the motor vehicle.

14. The system of claim 11, wherein the insurance is motor vehicle insurance.

15. The system of claim 11, wherein the insurance is health insurance.

16. The system of claim 11, wherein the insurance is life insurance.

17. The system of claim 11, wherein the duplicate driving data is recorded by the mobile device and by the personal monitoring device.

18. The system of claim 11, wherein the gaps in driving data are due to: failure of the mobile device components, personal monitoring device components or combinations thereof to capture the information; failure of the operator to carry the mobile device, personal monitoring device or combinations thereof; or combinations thereof.

19. A system for determining qualification for insurance, cost of insurance or combinations thereof, for a motor vehicle operator, comprising:

- (a) at least one mobile device,
 - (i) wherein the mobile device is not installed within the motor vehicle;
 - (ii) wherein the mobile device comprises hardware including a local memory, a battery, a Global Positioning System (GPS) component, and sensors; wherein the sensors include an accelerometer and a gyroscope; wherein the mobile device further provides functions including a GPS function and a communication function; and
 - (iii) wherein the mobile device is operative to at least collect and communicate information when the operator is operating the motor vehicle and when the operator is not operating the motor vehicle to determine qualification for insurance, cost of insurance or combinations thereof,
 - (a) wherein the information collected when the operator is operating the motor vehicle includes driving data; wherein the driving data is collected solely by the mobile device without receiving

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- such driving data from vehicle sensors, calculated or derived based on data solely collected by the mobile device without receiving such data from vehicle sensors, or combinations thereof; and
- (b) wherein the information collected when the operator is not operating the motor vehicle includes location; wherein at least a portion of the information collected when the operator is not operating the motor vehicle is used for determining qualification for insurance, cost of insurance or combinations thereof;
- (b) at least one application software resident on the mobile device,
- (i) wherein the application software is activated and deactivated to perform at least one of the said information collection and communication based on certain predetermined trigger events; and
- (ii) wherein the application software comprises a setup subsystem that is at least initiated when the user starts using the system; wherein the setup subsystem is operative to determine at least one trigger event used to activate or deactivate the application software;
- (c) at least one central control system that is operative to receive, process and communicate at least a portion of the information collected and/or derived by the mobile device for determining qualification for insurance, cost of insurance or combinations thereof;
- (d) at least one process to identify data issues in the information collected by the mobile device; wherein the information collected and processed for identifying data issues comprises driving data including date, time and speed collected by one or more components of the mobile device including GPS, accelerometer, or combinations thereof;
- wherein the data issues identified comprise:
- presence of insufficient driving data collected by one or more components of the mobile device; and
- presence of duplicate driving data, gaps in driving data, or combinations thereof recorded by the mobile device; and

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- (e) at least one algorithm, wherein the algorithm is operative to determine qualification for insurance, cost of insurance or combinations thereof based on at least a portion of the information collected by the mobile device when the operator is operating the motor vehicle and when the operator is not operating the motor vehicle.
20. The system of claim 19, further comprising at least one personal monitoring device,
- (a) wherein the personal monitoring device is a smart-watch; a fitness tracker; a portable medical, health and fitness device; or combinations thereof;
- (b) wherein the personal monitoring device includes one or more mechanisms for wearing the personal monitoring device and/or attaching the personal monitoring device to the operator when the operator is operating the motor vehicle and when the operating is not operating the motor vehicle; and
- (c) wherein the personal monitoring device is operative to at least collect and communicate additional information used for qualification for insurance, cost of insurance or combinations thereof for the operator, and wherein the information collected comprises accelerometer data, GPS data or combinations thereof.
21. The system of claim 19, wherein the mobile device is a smartphone.
22. The system of claim 19, wherein the insurance is motor vehicle insurance.
23. The system of claim 19, wherein the insurance is health insurance.
24. The system of claim 19, wherein the insurance is life insurance.
25. The system of claim 20, wherein the duplicate driving data is recorded by the personal monitoring device and by the mobile device.
26. The system of claim 19, wherein the gaps in driving data are due to: information not collected when the battery of the mobile device is being charged; an information gap when the operator is not carrying the mobile device; data errors due to low memory levels; data errors due to operating system failure of the mobile device; or a combination.

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